3-D UWB Anchor Autocalibration via MDS and Multilateration With Paired-Anchor Height Decoupling

Marios Raspopoulos, *Senior Member, IEEE*, and Iacovos Ioannou, *Senior Member, IEEE*

***Abstract*—Accurate anchor positions are required for ultra-wideband (UWB) indoor localization, yet manual surveying is impractical in opportunistic deployments. This article proposes a lightweight, purely anchor-based 3-D autocalibration framework for sparse multihop UWB networks requiring no mobile agents, inertial sensors, or learning components. Two algorithms are developed: an multidimensional scaling (MDS)-based approach using Floyd–Warshall gap-fill, Procrustes alignment and iterative pivot least squares refinement; and a multilateration-based greedy cascade. Both are evaluated in full 3-D and Semi-3D modes via a physics-based Monte Carlo simulator modeling realistic nonline-of-sight (NLOS) ranging errors. Standalone MDS and multilateration achieve approximately 1 m 3-D root mean squared error (RMSE), with height estimation identified as the dominant bottleneck. To address this, a paired-anchor deployment is introduced in which two anchors share each XY position with a fixed vertical separation. Through cross-pair ranging, a closed-form height estimator is obtained by which horizontal position error is canceled exactly. Results indicate that 3-D RMSE is reduced by up to 39% for MDS and 41% for multilateration relative to standalone baselines, reaching 0.54 and 0.53 m at $n_k = 7$ with negligible computational overhead. The cross-pair differential estimator further provides inherent NLOS/multipath resilience typical of UWB deployments. The reported results constitute a simulation-based proof-of-concept of the paired-anchor autocalibration approach and are intended to provide a quantitative foundation for subsequent experimental validation.**

***Index Terms*—3-D indoor localization, anchor autocalibration, multidimensional scaling (MDS), multilateration, ultra-wideband (UWB).**

I. Introduction

INDOOR localization has become a key enabling technology for a wide range of applications, including industrial automation, mobile robotics, asset tracking, emergency response, and smart environments [1]. Among the available radio-frequency technologies, ultra-wideband (UWB) has emerged as a leading solution due to its fine temporal resolution, robustness to multipath fading and ability to provide centimeter-level ranging precision [2], [3]. These properties render UWB particularly attractive for infrastructure-based indoor localization systems, where fixed reference nodes, known as anchors, are used to localize mobile tags.

A critical requirement for accurate UWB localization is the precise knowledge of anchor coordinates. Any error in anchor placement propagates directly into the estimated tag positions. In controlled deployments, anchors are typically surveyed manually using laser rangefinders or map-matching tools. However, in many real-world scenarios—such as emergency response, temporary industrial installations, or ad hoc deployments in unknown environments—manual surveying is impractical or impossible. Anchors may be placed opportunistically, under time pressure and without access to accurate surveying equipment. In such cases, the localization infrastructure itself must be capable of autonomously determining anchor positions from internode UWB measurements alone. This process is commonly referred to as *anchor autocalibration* or *self-calibration*.

Over the past two decades, a wide range of autocalibration techniques has been proposed, spanning optimization-based formulations, distributed algorithms, probabilistic inference and, more recently, multisensor fusion within simultaneous localization and mapping (SLAM) and factor-graph frameworks. While many of these methods achieve impressive accuracy under controlled conditions, their underlying assumptions limit applicability in large-scale or opportunistic deployments. Single-hop methods assume dense pairwise connectivity and typically operate in 2-D [4], [5]. Multihop methods relax the connectivity assumption but often rely on iterative global optimization [6] or probabilistic inference with substantial computational overhead [7]. Multisensor approaches achieve centimeter-level accuracy but require mobile robots, light detection and ranging (LiDAR), or inertial sensors that may not be available at the deployment site [8], [9], [10].

A further limitation shared by most existing methods is the assumption of 2-D deployment. In practice, UWB anchors are

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mounted at varying heights throughout a building to provide adequate 3-D coverage. The vertical component of anchor position is consistently the hardest to estimate from interanchor ranging alone, because anchors in typical indoor deployments span a narrow height range relative to their horizontal separation, creating a poorly conditioned vertical geometry. This limitation is largely overlooked in the literature, where 3-D autocalibration results are rare and height estimation accuracy is seldom reported independently.

This article addresses these limitations through two complementary contributions. The *first* is a pair of lightweight 3-D autocalibration algorithms—one based on multidimensional scaling (MDS) and one on multilateration—that operate exclusively on interanchor ranging measurements, tolerate missing links due to limited radio range or nonline-of-sight (NLOS) blockage and require no optimization solver, inertial sensors, or machine learning components. The MDS algorithm recovers the global anchor geometry via Floyd–Warshall gap-fill, classical MDS embedding, Procrustes alignment to the known-anchor frame and iterative pivot least squares (pivot-LS) refinement with a residual gate. The multilateration algorithm propagates the known-anchor set outward through the network via a confidence-ordered greedy cascade of pivot-LS solutions. Both operate in a full 3-D mode and a semi-3D mode that exploits inexpensively measured anchor heights as side information. The *second* contribution is a structured *paired-anchor* deployment strategy that places two anchors at each XY position with a fixed known vertical separation Δh. Cross-pair ranging measurements between a pair of unknown anchors and a pair of known references yield a *closed-form height estimator* that exactly cancels the horizontal interpair distance, making height estimation decoupled from horizontal position error for the first time in anchor autocalibration. This estimator, whose noise variance decreases as $1/\Delta h^2$, is applied as a postprocessing step to either backbone algorithm with negligible computational overhead.

The framework is evaluated via a physics-based Monte Carlo simulation platform that models UWB link budgets, building wall-crossing NLOS effects and distance-dependent ranging noise in a realistic multiroom indoor environment. Three independent parameter sweeps characterize the effect of known-anchor count, ranging noise level and network size on autocalibration accuracy. The paired-anchor algorithms are additionally evaluated across various vertical separations to quantify the accuracy–deployment-constraint tradeoff.

The main contributions of this work are as follows: 1) *MDS-based 3-D autocalibration* via Floyd–Warshall gap-fill, classical MDS embedding, Procrustes alignment and iterative pivot-LS refinement with a residual gate; 2) *multilateration-based 3-D autocalibration* via a greedy cascade of pivot-LS solutions; and 3) *paired-anchor deployment with a closed-form height estimator* derived from cross-pair differential ranging, by which the 3-D root mean squared error (RMSE) is significantly reduced relative to the standalone 3-D baseline.

The rest of this article is organized as follows. Section II reviews existing autocalibration methods and positions the proposed framework within this landscape. Section III formalizes the system model and the autocalibration problem. Section IV presents the MDS-based, multilateration-based and paired-anchor algorithms together with the simulation environment. Section V reports Monte Carlo evaluation results for both algorithm families across various parameter sweeps and discusses them in the context of related work. Section VI concludes this article and outlines future research directions.

II. Related Work

A. General Landscape

Anchor self-calibration methods can be broadly classified into *single-hop* and *multihop* approaches. Single-hop methods assume that all anchors can directly measure pairwise distances, forming a fully connected ranging graph. Under this assumption, calibration reduces to a constrained geometric reconstruction problem, typically solved using MDS, trilateration, optimization, or robust estimation.

Multihop methods operate on sparse and irregular connectivity graphs, where anchors may only measure distances to a subset of neighbors. Such conditions are common in large-scale deployments and in environments affected by NLOS propagation. Multihop calibration is closely related to cooperative localization in sensor networks, where nodes must infer their positions from incomplete and noisy internode measurements. These methods must address ambiguity, error accumulation and the absence of a global reference frame.

B. Single-Hop Self-Calibration

Early and representative single-hop approaches focus on robust and efficient reconstruction under dense connectivity. Random sample consensus-based minimal solvers for time-of-arrival calibration are presented in [11] and [12], achieving decimeter-level accuracy in small volumes. Time difference of arrival-based systems introduce clock asynchrony, addressed in [13] via joint synchronization and calibration. Optimization-based schemes refine anchor positions from rough initial estimates using a mobile tag [14], while iterative centroid and trilateration methods [15] demonstrate performance comparable to manual calibration.

A class of methods closely related to the approach proposed in this work is based on MDS. MDS reconstructs node coordinates from pairwise distances by solving a global distance-geometry problem [16]. While MDS-based approaches provide an elegant and computationally efficient way to recover a globally consistent geometry, they are primarily designed for relative positioning and assume sufficiently dense internode connectivity. Moreover, they do not explicitly address the challenges posed by indoor 3-D deployments, where vertical observability is weak and measurement graphs are often highly fragmented. These limitations become critical in opportunistic UWB deployments, motivating the multihop and paired-anchor extensions proposed in this work.

A probabilistic alternative to MDS and trilateration-based single-hop calibration is presented by Schwarzbach and Jung [7], who evaluated a grid-based Bayesian uncertainty propagation framework for collaborative UWB self-calibration in a fully meshed network of 12 nodes deployed in a $44 \times 30\,\mathrm{m}^2$ industrial

hall. Rather than collapsing anchor position uncertainty to a covariance matrix—as parametric approaches, such as extended Kalman filtering (EKF) do—the proposed method maintains and propagates discrete probability mass distributions across all nodes. During the measurement update, the likelihood of each grid cell is computed as a weighted superposition of contributions from the top-$\tau\%$ probability hypotheses of every reference node, thereby preserving multimodal and non-Gaussian uncertainty structure. The key practical advantage emerges under NLOS-contaminated initialization: while a closed-form baseline degrades catastrophically (median error 1.48–2.43 m, with extreme outliers exceeding 30 m), the grid-based method degrades gracefully (median error 0.62–0.99 m), with the mean-error ratio rising from 1.6× under LOS to 3.7–7.4× under NLOS initialization. The method also tolerates partial connectivity by incorporating only available reference nodes in the measurement update. These advantages come at the cost of significant computational overhead and a centralized architecture that requires aggregating all pairwise measurements at a single coordinator. Furthermore, the approach operates in 2-D and targets near-fully meshed networks, limiting applicability to sparse or large-scale 3-D deployments.

Distributed formulations further reduce centralization. In [4], anchors establish a coordinate system by fixing four reference nodes and minimizing a global cost over interanchor distances using gradient descent, achieving subdecimeter RMSE. Hybrid schemes combine optimization with learning-based error correction [5], or fuse UWB with inertial sensing in EKF frameworks [17]. While effective, these methods rely on dense connectivity and are therefore unsuitable for sparse or large-scale deployments.

C. Multihop and Network-Scale Methods

Multihop self-calibration relaxes the full connectivity assumption and targets large-scale networks. Classical solutions include iterative MDS [16], semidefinite programming relaxations [18], distributed self-positioning [19], nonparametric belief propagation [20], and simulated annealing [21]. These methods establish relative coordinate systems under incomplete distance information and scale to hundreds of nodes in simulation. A UWB multihop solution is presented in [6], combining linear least squares initialization with global optimization using limited-memory Broyden–Fletcher–Goldfarb–Shanno with box constraints (L-BFGS-B) and optional refinement via a mobile node. In an indoor environment with 18 anchors, a mean absolute error (MAE) of 21.6 cm is reported under severe NLOS. Although effective, this framework relies on heavy nonlinear optimization, requires mobile assistance in its complete form and is also limited to 2-D.

D. Recent Multisensor and SLAM-Based Approaches

Recent research increasingly integrates UWB with additional sensing modalities and SLAM-style optimization. Factor graph optimization frameworks jointly estimate anchor positions and robot trajectories using UWB and LiDAR [8]. Two-stage UWB–inertial measurement unit (IMU)–odometer pipelines employ MDS for initialization followed by graph-based refinement and learning-based NLOS mitigation [9]. SLAM-inspired inverse positioning [10] and Kalman-filter-based UWB–IMU fusion [22] achieve anchor RMSEs below 10 cm. System-level calibration using anchor-to-anchor and anchor-to-tag ranging with nonlinear least squares is reported in [23], substantially reducing ranging bias and deployment effort. These approaches demonstrate high accuracy and robustness, but at the cost of increased system complexity. They typically require mobile robots, inertial sensors, LiDAR, or predefined trajectories and rely on computationally intensive optimization.

E. Positioning of This Work

In contrast to the literature, this work proposes a *purely anchor-based*, lightweight framework that evaluates two complementary autocalibration strategies—MDS-based and multilateration-based—each operating independently in both full 3-D and semi-3D modes and extends both with a structured *paired-anchor* deployment that enables a closed-form, range-decoupled height estimator. By exploiting the vertical pair geometry to decouple Z estimation from horizontal ranging error—without any optimization solver, inertial sensor, or learning component—the proposed approach bridges the gap between graph-theoretic multihop localization and optimization-heavy, multisensor pipelines, making it well-suited for rapid, opportunistic deployments in sparse 3-D environments where only UWB infrastructure is available, such as emergency response and industrial installations.

III. System Model and Problem Description

An overview of the proposed 3-D UWB anchor autocalibration framework is shown in Fig. 1. The indoor environment and physical anchor deployment constitute the input. A physics-based channel model generates the sparse measured distance matrix $\widetilde{\mathbf{D}}$ from the link budget and ranging error models. The core algorithms recover an initial XY warm start together with a closed-form, height-decoupled estimate, and a final refinement stage produces the 3-D anchor positions. This section formalizes the system model and the autocalibration problem, while the algorithms themselves are presented in Section IV.

A. Network Topology and Problem Formulation

Consider a set of N UWB anchors deployed in a 3-D indoor environment. Let $\mathbf{p}_i = [x_i, y_i, z_i]^\top \in \mathbb{R}^3$ denote the true position of anchor i. The anchors are partitioned into two disjoint sets: the *known* set $\mathcal{K} \subset \{1, \ldots, N\}$, whose positions are assumed to be known with bounded uncertainty and the *unknown* set $\mathcal{U} = \{1, \ldots, N\} \setminus \mathcal{K}$, whose positions are to be recovered. The measured distance matrix $\widetilde{\mathbf{D}} \in \mathbb{R}^{N \times N}$ is sparse: entry $\widetilde{d}_{ij}$ is available only when anchors i and j share a radio link. Unavailable entries are represented as `NaN`. The *connectivity graph* $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has vertex set $\mathcal{V} = \{1, \ldots, N\}$ and edge set $\mathcal{E} = \{(i, j) : \widetilde{d}_{ij} \text{ is available}\}$. The autocalibration problem can be stated as

$$\hat{\mathbf{p}}_i = \arg\min_{\mathbf{p}_i} \sum_{(i,j)\in\mathcal{E}} w_{ij} \left(\|\mathbf{p}_i - \mathbf{p}_j\| - \widetilde{d}_{ij} \right)^2 \quad \forall\, i \in \mathcal{U} \tag{1}$$

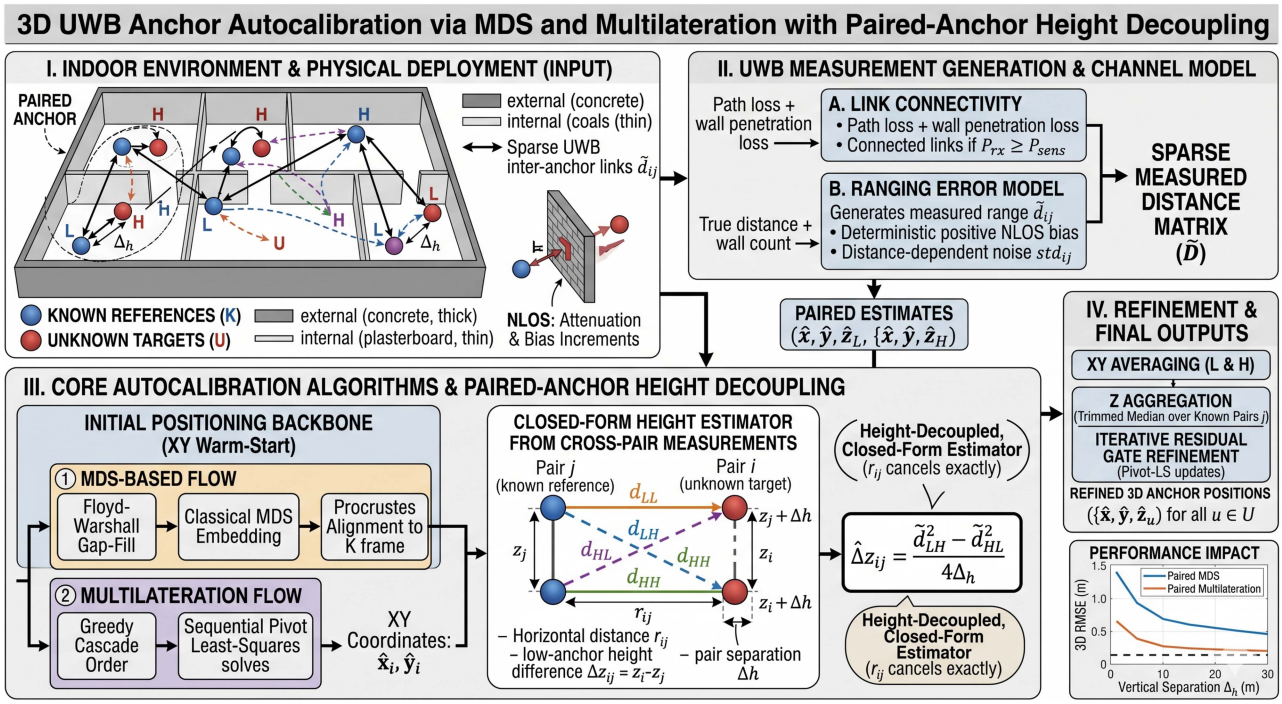


Fig. 1. Overview of the proposed 3-D UWB anchor autocalibration framework. (I) Indoor environment and physical anchor deployment, with paired low and high anchors separated by a fixed vertical offset Δh. (II) UWB measurement generation and channel model, comprising link connectivity and the ranging error model, yielding the sparse measured distance matrix $\widetilde{\mathbf{D}}$. (III) MDS-based and multilateration-based autocalibration algorithms providing the XY warm start, together with the closed-form paired-anchor height estimator derived from cross-pair measurements. (IV) Refinement stage producing the final 3-D anchor positions.

subject to the known-anchor constraints $\hat{\mathbf{p}}_i = \mathbf{p}_i + \boldsymbol{\eta}_i$ for all $i \in \mathcal{K}$, where $\boldsymbol{\eta}_i \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathrm{KA}}^2 \mathbf{I})$ models the positional measurement uncertainty of the known anchors with standard deviation σ_{KA}. The scalars $w_{ij} > 0$ are measurement reliability weights derived from the link budget, as detailed below.

B. UWB Link Budget and Connectivity Model

The received power at anchor j from a transmission by anchor i is computed using the standard log-distance path loss model augmented with wall penetration losses

$$P_{\mathrm{rx}}^{(ij)} = P_{\mathrm{tx}} - \underbrace{\mathrm{PL}_0 + 10\, n\, \log_{10}\left(\frac{d_{ij}}{d_0}\right)}_{\text{free-space path loss}} - \underbrace{n_{\mathrm{int}}^{(ij)} L_{\mathrm{int}} + n_{\mathrm{ext}}^{(ij)} L_{\mathrm{ext}}}_{\text{wall penetration loss}} \tag{2}$$

where P_{tx} is the transmit power in dBm, $\mathrm{PL}_0 = 20\log_{10}(4\pi d_0 f_c/c)$ is the free-space path loss at the reference distance $d_0 = 1$ m for carrier frequency f_c and speed of light c, n is the path loss exponent, $n_{\mathrm{int}}^{(ij)}$ and $n_{\mathrm{ext}}^{(ij)}$ are the numbers of internal and external wall intersections along the ray from i to j and L_{int}, and L_{ext} are the corresponding per-wall attenuation coefficients in dB. A link between anchors i and j exists only if the received power exceeds the receiver sensitivity threshold P_{sens}: $(i,j) \in \mathcal{E} \iff P_{\mathrm{rx}}^{(ij)} \geq P_{\mathrm{sens}}$. The wall intersection count $n_{\mathrm{int}}^{(ij)} + n_{\mathrm{ext}}^{(ij)}$ is computed geometrically by testing the segment $\overline{\mathbf{p}_i \mathbf{p}_j}$ against each facet of the building model.

C. Ranging Error Model

For each available link $(i,j) \in \mathcal{E}$, the measured range $\tilde{d}_{ij}$ deviates from the true Euclidean distance $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|$ due to multipath, NLOS propagation and hardware timing errors according to

$$\tilde{d}_{ij} = d_{ij} + b_{ij} + \varepsilon_{ij} \tag{3}$$

where b_{ij} is a deterministic positive NLOS bias and ε_{ij} is a zero-mean Gaussian noise term. Under LOS conditions ($n_{\mathrm{int}}^{(ij)} + n_{\mathrm{ext}}^{(ij)} = 0$): $b_{ij} = 0$ and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\mathrm{LOS}}^2)$. Under NLOS conditions, each wall crossing introduces additional bias and increases the noise variance

$$b_{ij} = n_{\mathrm{int}}^{(ij)} b_{\mathrm{int}} + n_{\mathrm{ext}}^{(ij)} b_{\mathrm{ext}} \tag{4}$$

$$\sigma_{ij} = \sigma_{\mathrm{NLOS}} + \left(n_{\mathrm{int}}^{(ij)} + n_{\mathrm{ext}}^{(ij)}\right) \sigma_{\mathrm{wall}} \tag{5}$$

where b_{int} and b_{ext} are the per-wall bias increments and σ_{wall} is the per-wall noise increment. The weight w_{ij} in (1) is defined as the inverse of the link variance: $w_{ij} = \sigma_{ij}^{-2}$.

D. Semi-3D Variant: Known Heights

In many practical indoor deployments, anchor heights can be measured inexpensively (e.g., with a tape measure or laser

rangefinder or a barometer) even when horizontal positions are unknown. We therefore consider a *semi-3D* operating mode in which the height z_i of every anchor is available as a noisy measurement

$$\tilde{z}_i = z_i + \delta_i, \quad \delta_i \sim \mathcal{N}\left(0, \sigma_Z^2\right) \tag{6}$$

with σ_Z characterizing the height measurement accuracy. In the semi-3D mode, the estimation problem reduces to recovering only the horizontal coordinates (x_i, y_i) for each $i \in \mathcal{U}$, with the vertical coordinate substituted by $\tilde{z}_i$ after every estimation step. This decoupling substantially improves accuracy in quasi-planar deployments where UWB has weak vertical observability, as anchors typically span a narrow height range relative to the room height.

IV. Methodology

This section presents the complete methodological framework of the proposed anchor autocalibration system. To support a rigorous and reproducible evaluation, we have developed a custom MATLAB-based simulation platform that synthesizes realistic UWB measurement data under configurable deployment conditions. The simulator allows the user to define the 3-D geometry of the indoor environment as well as the positions and heights of all UWB anchors. The radio front-end is characterized by user-defined parameters including transmit power, carrier frequency, path loss exponent, receiver sensitivity, and per-wall attenuation coefficients for both internal partitions and external structural walls. The channel model generates per-link received signal strength values from a log-distance path loss formula augmented with wall-crossing losses, determines link connectivity based on a received power threshold and produces noisy range measurements according to a distance-dependent LOS/NLOS error model with positive range bias and Gaussian noise whose variance increases with the number of wall intersections along each path. This simulation environment is used to drive a Monte Carlo evaluation framework in which anchor positions and known-anchor subsets are randomized across hundreds of independent trials, enabling statistically robust performance assessment under diverse deployment geometries and connectivity conditions.

Within this framework, two autocalibration algorithms are developed and evaluated, each in a standard full 3-D mode and a semi-3D variant that exploits known anchor heights. The first, *MDS-based autocalibration*, reconstructs the global anchor geometry from the incomplete distance matrix using MDS followed by iterative pivot-LS refinement. The second, *multilateration-based autocalibration*, estimates anchor positions sequentially via a greedy cascade of pivot-LS solutions, propagating the known set outward through the network. Both algorithms are then extended to a *paired-anchor* deployment in which anchors are installed in vertically separated pairs at each XY position. This structured deployment enables a closed-form, range-decoupled height estimator that substantially reduces vertical localization error while preserving the horizontal accuracy of the underlying algorithm.

A. MDS-Based Autocalibration

Classical MDS recovers a Euclidean embedding from a complete matrix of pairwise squared distances. Applying MDS directly to $\widetilde{\mathbf{D}}$ is problematic because: 1) many entries are missing due to limited radio range; 2) measurement noise violates the Euclidean distance constraint; and 3) the solution is defined only up to a rigid-body transformation. We address each of these challenges in turn.

1) Floyd–Warshall Gap-Fill: Missing entries in $\widetilde{\mathbf{D}}$ are imputed using all-pairs shortest paths computed by the Floyd–Warshall algorithm [24], [25]. Defining the initialization $\mathbf{F}^{(0)}$ with $F_{ij}^{(0)} = \widetilde{d}_{ij}$ for $(i,j) \in \mathcal{E}$, $F_{ii}^{(0)} = 0$ and $F_{ij}^{(0)} = \infty$ otherwise, the algorithm iterates

$$F_{ij}^{(k)} = \min\left(F_{ij}^{(k-1)},\ F_{ik}^{(k-1)} + F_{kj}^{(k-1)}\right), \quad k = 1, \ldots, N \tag{7}$$

for all i and j. Entries that were originally missing and have been filled by a shortest path of length $\ell \geq 2$ hops are scaled by a shrinkage factor $\alpha \in (0,1)$

$$\hat{F}_{ij} = \begin{cases} \widetilde{d}_{ij}, & (i,j) \in \mathcal{E} \\ \alpha\, F_{ij}^{(N)}, & (i,j) \notin \mathcal{E} \end{cases} \tag{8}$$

yielding the imputed matrix $\hat{\mathbf{F}}$. The shrinkage corrects the systematic overestimation introduced by multihop path routing, which accumulates intermediate ranging noise along the path. The resulting symmetric matrix $\hat{\mathbf{F}}$ is made symmetric by averaging: $\hat{F}_{ij} \leftarrow (\hat{F}_{ij} + \hat{F}_{ji})/2$.

2) Classical MDS Embedding: Given the imputed squared-distance matrix $\hat{\mathbf{F}}^{\circ 2}$, the double-centring operator produces the Gram matrix

$$\mathbf{B} = -\tfrac{1}{2}\,\mathbf{J}\,\hat{\mathbf{F}}^{\circ 2}\,\mathbf{J} \tag{9}$$

where $\mathbf{J} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ is the centring matrix. If the imputed distances were Euclidean, then $\mathbf{B}$ would equal $\mathbf{X}\mathbf{X}^\top$ for some coordinate matrix $\mathbf{X}$. In practice, $\mathbf{B}$ is symmetrized and its eigenvalue decomposition is computed as $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, where the columns of $\mathbf{V}$ are unit eigenvectors and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$. Retaining the top three eigenpairs $(\lambda_k, \mathbf{v}_k)$, $k = 1, 2, 3$, the 3-D embedding is

$$\hat{\mathbf{X}} = \mathbf{V}_{1:3}\,\operatorname{diag}\left(\sqrt{\max(\lambda_1, 0)},\ \sqrt{\max(\lambda_2, 0)},\ \sqrt{\max(\lambda_3, 0)}\right) \tag{10}$$

where negative eigenvalues (arising from non-Euclidean imputed distances) are clamped to zero. The fraction of total variance explained by the 3-D embedding is $\sum_{k=1}^{3} \max(\lambda_k, 0) / \sum_{k=1}^{N} \max(\lambda_k, 0)$, which serves as an indicator of embedding quality.

3) Procrustes Alignment: The embedding $\hat{\mathbf{X}}$ is defined in an arbitrary coordinate frame. It must be aligned to the global frame defined by the known anchors $\{\mathbf{p}_i : i \in \mathcal{K}\}$. Let $\mathbf{P} = \hat{\mathbf{X}}_{\mathcal{K}} \in \mathbb{R}^{K \times 3}$ be the embedded positions of the known anchors and $\mathbf{Q} \in \mathbb{R}^{K \times 3}$ be their true (noisy) positions. The orthogonal Procrustes problem [26], [27] seeks the rotation matrix $\mathbf{R} \in$

SO(3) and translation vector $\mathbf{t} \in \mathbb{R}^3$ that minimize

$$\min_{\mathbf{R},\mathbf{t}} \left\|\mathbf{P}\mathbf{R}^\top + \mathbf{1}\mathbf{t}^\top - \mathbf{Q}\right\|_F^2. \tag{11}$$

Denoting the centered matrices $\bar{\mathbf{P}} = \mathbf{P} - \bar{\mathbf{p}}\mathbf{1}^\top$ and $\bar{\mathbf{Q}} = \mathbf{Q} - \bar{\mathbf{q}}\mathbf{1}^\top$, the solution is obtained via singular value decomposition (SVD) of the cross-covariance matrix $\mathbf{M} = \bar{\mathbf{P}}^\top\bar{\mathbf{Q}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$

$$\mathbf{R} = \mathbf{V}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & \det(\mathbf{V}\mathbf{U}^\top)\end{bmatrix}\mathbf{U}^\top, \qquad \mathbf{t} = \bar{\mathbf{q}} - \bar{\mathbf{p}}\,\mathbf{R}^\top \tag{12}$$

where the determinant term corrects for improper rotations (reflections), ensuring $\det(\mathbf{R}) = +1$. The aligned embedding is $\hat{\mathbf{X}}^* = \hat{\mathbf{X}}\mathbf{R}^\top + \mathbf{1}\mathbf{t}^\top$. The Procrustes residual $\varepsilon_{\text{proc}} = \|\hat{\mathbf{X}}^*_{\mathcal{K}} - \mathbf{Q}\|_F / \sqrt{K}$ quantifies the alignment quality and is used as a proxy for embedding accuracy in the residual-gated refinement stage.

4) Iterative Pivot-LS Refinement With Residual Gate: The MDS embedding provides a warm start for each unknown anchor. Each anchor position is then refined iteratively using a weighted pivot-LS procedure. For anchor $u \in \mathcal{U}$, let $\mathcal{R}_u = \{j : (u,j) \in \mathcal{E},\ \hat{\mathbf{p}}_j \text{ available}\}$ denote its reference set (known or already estimated anchors with measured links). Starting from (3) and squaring

$$\|\mathbf{p}_u - \mathbf{p}_j\|^2 = \widetilde{d}_{uj}^2 + \text{error terms}. \tag{13}$$

Selecting anchor $r_1 \in \mathcal{R}_u$ as the pivot and subtracting its equation from that of each $r_j \in \mathcal{R}_u \setminus \{r_1\}$ eliminates the unknown $\|\mathbf{p}_u\|^2$, yielding the linear system

$$\mathbf{A}\begin{bmatrix}x_u\\ y_u\end{bmatrix} = \mathbf{b} \tag{14}$$

where for $j = 2, \ldots, |\mathcal{R}_u|$

$$\mathbf{A} = 2\left[\mathbf{p}_{r_j,\text{xy}} - \mathbf{p}_{r_1,\text{xy}}\right] \tag{15}$$

$$b_j = d_{\text{xy},r_j}^2 - d_{\text{xy},r_1}^2 + \|\mathbf{p}_{r_1,\text{xy}}\|^2 - \|\mathbf{p}_{r_j,\text{xy}}\|^2. \tag{16}$$

Here, $d_{\text{xy},r_j} = \sqrt{\max(0,\ \widetilde{d}_{ur_j}^2 - (z_u - z_{r_j})^2)}$ is the projected horizontal range after accounting for the known or estimated height difference. The jth row of $\mathbf{A}$ and the scalar b_j are weighted by $w_j = 1/\max(d_{\text{xy},r_j}, \epsilon)$ to downweight references that are nearly coplanar with u in the horizontal plane. The system is solved in the minimum-norm least squares sense

$$\begin{bmatrix}\hat{x}_u\\ \hat{y}_u\end{bmatrix} = (\mathbf{W}^{1/2}\mathbf{A})^\dagger\,\mathbf{W}^{1/2}\mathbf{b} \tag{17}$$

where $\mathbf{W} = \text{diag}(w_2, \ldots, w_{|\mathcal{R}_u|})$ and $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse.

In *3-D mode*, the height z_u is estimated by inverse-distance weighting (IDW) over the known anchors $\mathcal{K}_u = \mathcal{K} \cap \mathcal{R}_u$

$$\hat{z}_u = \begin{cases}\dfrac{\sum_{k\in\mathcal{K}_u} \widetilde{d}_{UK}^{-1} z_k}{\sum_{k\in\mathcal{K}_u} \widetilde{d}_{UK}^{-1}}, & \mathcal{K}_u \neq \emptyset\\ \bar{z}_{\mathcal{K}}, & \mathcal{K}_u = \emptyset\end{cases} \tag{18}$$

Algorithm 1: MDS Autocalibration.

Require: $\widetilde{\mathbf{D}}$, $\mathcal{K}$, $\{\mathbf{p}_i\}_{i\in\mathcal{K}}$, $\{\tilde{z}_i\}_{i=1}^N$ (Semi-3D only), α, P
Ensure: $\{\hat{\mathbf{p}}_i\}_{i\in\mathcal{U}}$
Compute $\hat{\mathbf{F}}$ via Floyd-Warshall with shrinkage α (7)–(8)
Compute Gram matrix $\mathbf{B}$ (9)
Eigendecompose $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$; form $\hat{\mathbf{X}}$ (10)
Compute $\mathbf{R}$, $\mathbf{t}$ via Procrustes; align $\hat{\mathbf{X}}^* \leftarrow \hat{\mathbf{X}}\mathbf{R}^\top + \mathbf{1}\mathbf{t}^\top$ (12)
$\hat{\mathbf{p}}_u \leftarrow \text{clip}(\hat{\mathbf{X}}^*_u, \mathbf{0}, \mathbf{r}_{\max})$ for all $u \in \mathcal{U}$
if Semi-3D **then** $\hat{z}_u \leftarrow \tilde{z}_u$ for all $u \in \mathcal{U}$
end if
for $p = 1, \ldots, P$ **do**
  Sort $\mathcal{U}$ by $|\{(u,j) \in \mathcal{E}\}|$ descending
  for each $u \in \mathcal{U}$ (sorted) **do**
    Compute $\hat{z}_u$ via IDW (18) (3D) or $\hat{z}_u \leftarrow \tilde{z}_u$ (Semi-3D)
    Solve (17) to get $\hat{\mathbf{p}}_u^{\text{new}}$
    Apply residual gate (20)
  end for
end for
return $\{\hat{\mathbf{p}}_i\}_{i\in\mathcal{U}}$

where $\bar{z}_{\mathcal{K}}$ is the mean height of all known anchors. In the *semi-3D mode*, $\hat{z}_u = \tilde{z}_u$ from (6) is substituted directly, bypassing the IDW estimation entirely. A *residual gate* controls the acceptance of each refined estimate. Let $\rho(\mathbf{p}, \mathcal{R}_u)$ denote the root mean squared ranging residual of position $\mathbf{p}$ against the reference set $\mathcal{R}_u$

$$\rho(\mathbf{p}, \mathcal{R}_u) = \sqrt{\frac{1}{|\mathcal{R}_u|}\sum_{j\in\mathcal{R}_u}\left(\|\mathbf{p} - \hat{\mathbf{p}}_j\| - \widetilde{d}_{uj}\right)^2}. \tag{19}$$

The refined position $\hat{\mathbf{p}}_u^{\text{new}}$ is accepted only if it reduces the residual relative to the current estimate $\hat{\mathbf{p}}_u^{\text{old}}$

$$\hat{\mathbf{p}}_u \leftarrow \begin{cases}\hat{\mathbf{p}}_u^{\text{new}}, & \rho\left(\hat{\mathbf{p}}_u^{\text{new}}, \mathcal{R}_u\right) < \rho\left(\hat{\mathbf{p}}_u^{\text{old}}, \mathcal{R}_u\right)\\ \hat{\mathbf{p}}_u^{\text{old}}, & \text{otherwise.}\end{cases} \tag{20}$$

This gate prevents poorly estimated neighbors from degrading a well-placed anchor through successive refinement passes. Anchors are processed in descending order of link count to maximize the number of reliable references available at each step. The process is repeated for P passes (typically $P = 3$). Algorithm 1 summarizes the MDS autocalibration procedure.

B. Multilateration-Based Autocalibration

While MDS recovers the global geometric structure simultaneously, it requires a sufficiently dense and connected measurement graph to produce a reliable embedding. Multilateration takes the complementary approach: it estimates one anchor at a time from locally available measurements, propagating the known set outwards through the network via a greedy cascade. This makes multilateration resilient to fragmented connectivity, as each anchor only requires a small number of reliable neighbors rather than a globally well-connected graph.

1) Greedy Cascade Warm Start: The multilateration autocalibration begins with a greedy cascade that builds the reference set incrementally. At each step, the unresolved anchor with the most available references (known or already estimated anchors connected by a measured link) is selected for estimation. Ties are broken by preferring the anchor with the smallest mean distance to its references, as closer references typically yield better conditioned geometry. Formally, the selection criterion at iteration t is

$$u^* = \arg\max_{u \in \mathcal{U}^{(t)}} \left| \mathcal{R}_u^{(t)} \right| \tag{21}$$

with tie-breaking on $\bar{d}_u = |\mathcal{R}_u^{(t)}|^{-1} \sum_{j \in \mathcal{R}_u^{(t)}} \widetilde{d}_{uj}$. The selected anchor u^* is immediately estimated using (17) and promoted to the reference set: $\mathcal{K}^{(t+1)} = \mathcal{K}^{(t)} \cup \{u^*\}$. The cascade continues until either all anchors are estimated or no unresolved anchor has $|\mathcal{R}_u^{(t)}| \geq M_{\min}$, where $M_{\min} \geq 3$ is the minimum number of references required.

The greedy cascade differs from MDS-based initialization in a key respect: each anchor is estimated using the *most recently updated* reference positions rather than a globally computed embedding. This means that errors in early estimates can propagate to later ones—the cascade order critically affects accuracy. The strategy of processing the most connected anchor first mitigates this by building the reference set from anchors with the strongest geometric constraints before attempting anchors with fewer or more distant links.

2) Pivot-LS Formulation and Z-Estimation: The pivot-LS formulation follows exactly as described in Section IV-A through (17). The key difference in the multilateration context is that the reference set $\mathcal{R}_u$ includes both originally known anchors and anchors estimated by the cascade, rather than only the MDS-aligned positions. This allows the cascade to reach anchors that have no direct link to any originally known node, provided a path exists through the network.

Height estimation in 3-D mode follows (18), restricted to known anchors in $\mathcal{K}_u$: only originally known anchors (not cascade estimates) contribute to the IDW average for z_u, as their heights are known with bounded uncertainty σ_{KA}, whereas cascade-estimated heights may carry accumulated error. In semi-3D mode, $\hat{z}_u = \tilde{z}_u$ regardless of the reference set composition.

3) Iterative Residual Gate Refinement: Following the greedy cascade, all estimated positions are refined using the same iterative residual gate procedure described in Section IV-A. Crucially, at this stage the reference set for each anchor u includes *all* anchors with finite estimated positions and a measured link to u—not only originally known ones—since the cascade estimates have been stabilized by the greedy ordering. This enables mutual refinement between adjacent estimated anchors, progressively reducing the accumulated cascade error. Algorithm 2 summarizes the full procedure.

Algorithm 2: Multilateration Autocalibration.

Require: $\widetilde{\mathbf{D}}$, $\mathcal{K}$, $\{\mathbf{p}_i\}_{i \in \mathcal{K}}$, $\{\tilde{z}_i\}_{i=1}^N$ (Semi-3D only), $M_{\min}$, P
Ensure: $\{\hat{\mathbf{p}}_i\}_{i \in \mathcal{U}}$
$\mathcal{K}^{(1)} \leftarrow \mathcal{K}$;; $\mathcal{U}^{(1)} \leftarrow \mathcal{U}$;; $t \leftarrow 1$
while $\mathcal{U}^{(t)} \neq \emptyset$ **do**
  Select u^* via (21)
  if no $u \in \mathcal{U}^{(t)}$ has $|\mathcal{R}_u^{(t)}| \geq M_{\min}$ **then break**
  end if
  Estimate $\hat{z}_{u^*}$ via IDW (18) (3D) or $\leftarrow \tilde{z}_{u^*}$ (Semi-3D)
  Solve (17) to get $\hat{\mathbf{p}}_{u^*}$
  $\mathcal{K}^{(t+1)} \leftarrow \mathcal{K}^{(t)} \cup \{u^*\}$;; $\mathcal{U}^{(t+1)} \leftarrow \mathcal{U}^{(t)} \setminus \{u^*\}$;; $t \leftarrow t+1$
end while
for $p = 1, \ldots, P$ **do** ▷Gate refinement
  Sort $\mathcal{U}$ by $|\{(u,j) \in \mathcal{E}\}|$ descending
  for each $u \in \mathcal{U}$ **do**
    Compute $\hat{z}_u$ (3D) or use $\tilde{z}_u$ (Semi-3D)
    Solve (17); apply gate (20)
  end for
end for
return $\{\hat{\mathbf{p}}_i\}_{i \in \mathcal{U}}$

C. Paired-Anchor Autocalibration

The standalone MDS and multilateration methods described in Sections IV-A and IV-B treat each anchor independently and rely on the IDW average (18) to recover height, which is sensitive to ranging noise in the vertical component. We propose a structured *paired-anchor* deployment that breaks the height–range ambiguity exactly, yielding a closed-form height estimate that is decoupled from horizontal position error.

1) Deployment Model and Notation: Instead of placing N anchors at arbitrary 3-D locations, anchors are deployed in pairs: each of $M = N/2$ distinct XY positions hosts one *low* anchor L_i at height z_i and one *high* anchor H_i at height $z_i + \Delta h$, where $\Delta h > 0$ is a fixed, known vertical separation shared by all pairs. Let $\mathcal{M} = \{1, \ldots, M\}$ index the XY positions and let $\text{pm} \in \mathbb{Z}^{M \times 2}$ be the pair-membership matrix whose row i contains the node indices $[L_i, H_i]$.

A subset $\mathcal{K}_p \subset \mathcal{M}$, $|\mathcal{K}_p| = n_k$, denotes the *known pairs:* positions for which both L_i and H_i are calibrated references with positions known to within uncertainty σ_{KA}. The remaining $M - n_k$ positions form the unknown set $\mathcal{U}_p = \mathcal{M} \setminus \mathcal{K}_p$. Intrapair links (between L_i and H_i at the same XY position) are excluded from the range matrix $\widetilde{\mathbf{D}}$, as their short separation amplifies noise in the differential formulas derived below.

2) Closed-Form Height Estimation From Cross-Pair Measurements: Consider unknown pair $i \in \mathcal{U}_p$ and any reference pair $j \in \mathcal{K}_p$. Let r_{ij} denote the horizontal distance between the two XY positions and $\Delta z_{ij} = z_i - z_j$ the height difference between their low anchors. The four interpair ranges satisfy

$$d_{LL,ij}^2 = r_{ij}^2 + \Delta z_{ij}^2 \tag{22}$$

$$d_{LH,ij}^2 = r_{ij}^2 + \left(\Delta z_{ij} + \Delta h\right)^2 \tag{23}$$

$$d_{HL,ij}^2 = r_{ij}^2 + \left(\Delta z_{ij} - \Delta h\right)^2 \tag{24}$$

$$d_{HH,ij}^2 = r_{ij}^2 + \Delta z_{ij}^2. \tag{25}$$

Subtracting (24) from (23) and expanding

$$d_{LH,ij}^2 - d_{HL,ij}^2 = \left(\Delta z_{ij} + \Delta h\right)^2 - \left(\Delta z_{ij} - \Delta h\right)^2 = 4\,\Delta z_{ij}\,\Delta h. \tag{26}$$

Solving for the height difference gives the *closed-form estimator*

$$\widehat{\Delta z}_{ij} = \frac{\tilde{d}_{LH,ij}^2 - \tilde{d}_{HL,ij}^2}{4\,\Delta h} \tag{27}$$

where $\tilde{d}_{LH,ij}$ and $\tilde{d}_{HL,ij}$ are the measured ranges. Note that r_{ij} cancels exactly in (26), so $\widehat{\Delta z}_{ij}$ is independent of horizontal position error. The estimated height of unknown pair i is then

$$\hat{z}_i = z_j + \widehat{\Delta z}_{ij} \tag{28}$$

for each reference pair j that provides a valid estimate. Multiple reference pairs yield multiple estimates; a 20% trimmed median is used to aggregate them robustly

$$\hat{z}_i = \mathrm{TrimMed}_{0.20}\left(\left\{z_j + \widehat{\Delta z}_{ij}\right\}_{j\in\mathcal{K}_p}\right). \tag{29}$$

The noise sensitivity of (27) is governed by the ranging noise variance σ^2 and the separation Δh. Linearising around the true ranges

$$\mathrm{Var}\left(\widehat{\Delta z}_{ij}\right) \approx \frac{d_{LH,ij}^2 + d_{HL,ij}^2}{8\,\Delta h^2}\,\sigma^2 \tag{30}$$

which decreases quadratically with Δh, providing the principal motivation for maximizing the vertical separation within the available ceiling height.

Once $\widehat{\Delta z}_{ij}$ is available, the horizontal interpair distance is recovered from (22)

$$\hat{r}_{ij} = \sqrt{\max\left(0,\ \tilde{d}_{LL,ij}^2 - \widehat{\Delta z}_{ij}^2\right)}. \tag{31}$$

These two quantities, $\widehat{\Delta z}_{ij}$ and $\hat{r}_{ij}$, are assembled into pair-level matrices $\mathbf{\Delta Z}$ and $\mathbf{R}_{xy}$ by the CROSSPAIRDIST subroutine (see Algorithm 3).

3) Paired-Anchor MDS Autocalibration: The paired MDS algorithm (see Algorithm 4) applies the original 3-D MDS pipeline of Section IV-A to all N individual anchors, treating the low and high anchor of each known pair as two independent references at their respective 3-D positions. Intrapair links are set to NaN so that they do not contaminate the Floyd–Warshall embedding. The pair structure enters only in the postprocessing step, where it provides two independent position estimates for each unknown XY location.

Because the same horizontal position is estimated by both $\hat{\mathbf{p}}_{L_i}$ and $\hat{\mathbf{p}}_{H_i}$, their XY components are averaged

$$\hat{\mathbf{x}}_i = \tfrac{1}{2}\left(\hat{\mathbf{p}}_{L_i}^{(xy)} + \hat{\mathbf{p}}_{H_i}^{(xy)}\right) \tag{32}$$

which reduces the XY variance by a factor of two relative to either anchor alone, provided the two MDS estimates are approximately independent.

The height of pair i is estimated by applying (27)–(29) with reference pairs $j \in \mathcal{K}_p$, using the range measurements directly from $\widetilde{\mathbf{D}}$. This replaces the IDW estimate (18) with the closed-form differential formula, which is immune to the horizontal position error that would otherwise propagate into the IDW weights.

Algorithm 3: CROSSPAIRDIST: Cross-Pair Distance Matrices.

Require: Range matrix $\widetilde{\mathbf{D}} \in \mathbb{R}^{N\times N}$, pair-membership pm $\in \mathbb{Z}^{M\times 2}$, vertical separation Δh
Ensure: $\mathbf{R}_{xy} \in \mathbb{R}^{M\times M}$, $\mathbf{\Delta Z} \in \mathbb{R}^{M\times M}$
Initialise $\mathbf{R}_{xy} \leftarrow \mathbf{NaN}$,; $\mathbf{\Delta Z} \leftarrow \mathbf{NaN}$
for each pair (i, j) with $i \neq j$ **do**
  $L_i, H_i \leftarrow \mathrm{pm}(i,:)$;; $L_j, H_j \leftarrow \mathrm{pm}(j,:)$
  Read $\tilde{d}_{LH} \leftarrow \widetilde{D}_{L_i,H_j}$,; $\tilde{d}_{HL} \leftarrow \widetilde{D}_{H_i,L_j}$,; $\tilde{d}_{LL} \leftarrow \widetilde{D}_{L_i,L_j}$
  if both $\tilde{d}_{LH}$ and $\tilde{d}_{HL}$ are valid **then**
    $\widehat{\Delta Z}_{ij} \leftarrow (\tilde{d}_{LH}^2 - \tilde{d}_{HL}^2)\ /\ (4\,\Delta h)$ ▷(27)
    $R_{xy,ij} \leftarrow \sqrt{\max(0,\ \tilde{d}_{LL}^2 - \widehat{\Delta Z}_{ij}^2)}$ ▷(31)
  else if $\tilde{d}_{LL}$ is valid **then**
    $\widehat{\Delta Z}_{ij} \leftarrow$ NaN;; $R_{xy,ij} \leftarrow \tilde{d}_{LL}$ ▷single-link fallback
  end if
end for
return $\mathbf{R}_{xy}$, $\mathbf{\Delta Z}$

Algorithm 4: Paired-Anchor MDS Autocalibration.

Require: $\widetilde{\mathbf{D}}$, pm, $\mathcal{K}_p$, known positions $\{\mathbf{p}_j\}_{j\in\mathcal{K}_p}$, Δh, mode, parameters Θ
Ensure: $\{\hat{\mathbf{x}}_i, \hat{z}_i\}_{i\in\mathcal{U}_p}$
Build individual-anchor reference set: $\mathcal{K}_{\text{ind}} \leftarrow \{L_j, H_j : j \in \mathcal{K}_p\}$
Set $\widetilde{D}_{L_i,H_i} \leftarrow$ NaN and $\widetilde{D}_{H_i,L_i} \leftarrow$ NaN for all $i \in \mathcal{M}$ ▷exclude intra-pair links
Run the MDS pipeline (Algorithm 1) on $\widetilde{\mathbf{D}}$ with references $\mathcal{K}_{\text{ind}} \to \hat{\mathbf{P}} \in \mathbb{R}^{N\times 3}$
for each unknown pair $i \in \mathcal{U}_p$ **do**
  $\hat{\mathbf{x}}_i \leftarrow \frac{1}{2}\left(\hat{\mathbf{P}}_{L_i}^{(xy)} + \hat{\mathbf{P}}_{H_i}^{(xy)}\right)$ ▷(32)
  $\hat{\mathbf{x}}_i \leftarrow \mathrm{clip}(\hat{\mathbf{x}}_i,\ [0, \ell_x] \times [0, \ell_y])$
  Compute $\{\mathbf{\Delta Z}, \mathbf{R}_{xy}\} \leftarrow$ CROSSPAIRDIST$(\widetilde{\mathbf{D}}, \mathrm{pm}, \Delta h)$
  $\hat{z}_i \leftarrow \mathrm{TrimMed}_{0.20}\left(\{z_j + \Delta Z_{ij}\}_{j\in\mathcal{K}_p,\ \Delta Z_{ij}\neq\text{NaN}}\right)$ ▷(29)
  if no valid ΔZ_{ij} exists **then**
    $\hat{z}_i \leftarrow \frac{1}{2}\left(\hat{P}_{L_i,z} + \hat{P}_{H_i,z} - \Delta h\right)$ ▷MDS fallback
  end if
  $\hat{z}_i \leftarrow \mathrm{clip}(\hat{z}_i,\ [0,\ \ell_z - \Delta h])$
end for
return $\{\hat{\mathbf{x}}_i, \hat{z}_i\}$

4) Paired-Anchor Multilateration Autocalibration: The paired multilateration algorithm (see Algorithm 5) follows an identical postprocessing strategy but replaces the MDS backbone with the greedy cascade of Section IV-B. The original 3-D multilateration runs on all N individual anchors; the pair structure is again used only in postprocessing.

Algorithm 5: Paired-Anchor Multilateration Autocalibration.

Require: $\widetilde{\mathbf{D}}$, pm, $\mathcal{K}_p$, known positions $\{\mathbf{p}_j\}_{j\in\mathcal{K}_p}$, Δh, mode, parameters Θ
Ensure: $\{\hat{\mathbf{x}}_i, \hat{z}_i\}_{i\in\mathcal{U}_p}$

Build $\mathcal{K}_{\text{ind}} \leftarrow \{L_j, H_j : j \in \mathcal{K}_p\}$
Set $\widetilde{D}_{L_i,H_i} \leftarrow$ NaN for all $i \in \mathcal{M}$ ▷exclude intra-pair links
Run multilateration cascade (Alg. 2) on $\widetilde{\mathbf{D}}$ with references $\mathcal{K}_{\text{ind}} \rightarrow \hat{\mathbf{P}} \in \mathbb{R}^{N\times 3}$
for each unknown pair $i \in \mathcal{U}_p$ **do**
  if both $\hat{\mathbf{P}}_{L_i}$ and $\hat{\mathbf{P}}_{H_i}$ valid **then**
    $\hat{\mathbf{x}}_i \leftarrow \frac{1}{2}\big(\hat{\mathbf{P}}^{(xy)}_{L_i} + \hat{\mathbf{P}}^{(xy)}_{H_i}\big)$ ▷(32)
  else if only one of $\hat{\mathbf{P}}_{L_i}$, $\hat{\mathbf{P}}_{H_i}$ valid **then**
    $\hat{\mathbf{x}}_i \leftarrow$ XY of the valid estimate
  else
    $\hat{\mathbf{x}}_i \leftarrow [\ell_x/2, \ell_y/2]$ ▷room-centre fallback
  end if
  $\hat{\mathbf{x}}_i \leftarrow \text{clip}(\hat{\mathbf{x}}_i, [0,\ell_x]\times[0,\ell_y])$
  Compute $\{\boldsymbol{\Delta}\mathbf{Z}, \mathbf{R}_{xy}\} \leftarrow \textsc{CrossPairDist}(\widetilde{\mathbf{D}}, \text{pm}, \Delta h)$
  $\hat{z}_i \leftarrow \text{TrimMed}_{0.20}\big(\{z_j + \Delta Z_{ij}\}_{j\in\mathcal{K}_p,\, \Delta Z_{ij}\neq\text{NaN}}\big)$ ▷(29)
  if no valid ΔZ_{ij} exists **then**
    $\hat{z}_i \leftarrow$ IDW average of $\hat{P}_{L_i,z}$ and $\hat{P}_{H_i,z} - \Delta h$ ▷multilat fallback
  end if
  $\hat{z}_i \leftarrow \text{clip}(\hat{z}_i, [0, \ell_z - \Delta h])$
end for
return $\{\hat{\mathbf{x}}_i, \hat{z}_i\}$

Multilateration is particularly suited to the paired setting because the cascade can exploit the doubled reference count: both anchors of each known pair enter $\mathcal{K}_{\text{ind}}$, providing more well-placed references and improving the conditioning of the pivot-LS system, especially for unknown anchors that have limited links to known nodes.

Table I lists all configurable parameters. The paired approach introduces two new parameters: the vertical separation Δh and the number of known pairs n_k.

TABLE I
ALGORITHM PARAMETERS AND RECOMMENDED VALUES

Symbol	Description	Value	Eq.
α	FW shrinkage factor	0.90	(8)
P	Gate refinement passes	3	(20)
$M_{\min}$	Min references (multilat)	3	(21)
M_{target}	Reference saturation count	3	—
Δh	Vertical pair separation	0.5–2.0 m	(27)
n_k	Known pairs	3–7	(28)
σ_{res}	Residual scale (multilat)	0.55 m	—
σ_{res}	Residual scale (MDS)	1.20 m	—
γ	Geometry condition scale	10	—
P_{weak}	RSSI lower bound	-100 dBm	—
P_{strong}	RSSI upper bound	-60 dBm	—
δ_{room}	Room bound margin	1.0 m	—

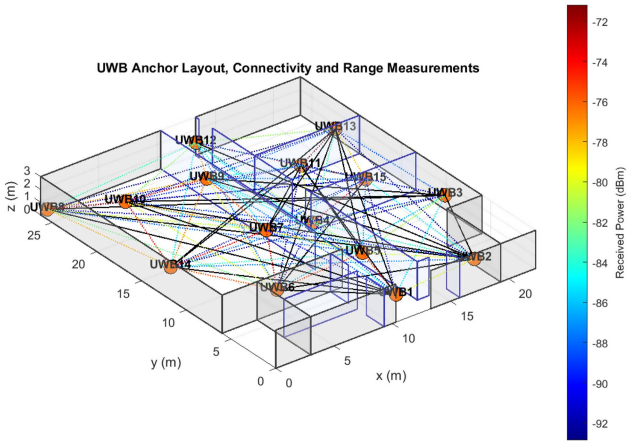


Fig. 2. Illustrative UWB anchor deployment ($N = 15$).

V. RESULTS

A. Environment and UWB Simulation Model

All evaluations in this section are conducted using the custom physics-based MATLAB simulator introduced in Section III. This section describes the building geometry, UWB radio parameters and measurement generation procedure used throughout the Monte Carlo studies.

1) Building Geometry: The simulated environment represents a multiroom indoor floor plan with overall dimensions 22.3 m × 26.3 m × 3.0 m, modeled as a set of 21 vertical planar facets. Seven facets represent load-bearing external walls (reinforced concrete) and 14 represent internal partitions (plasterboard). The layout includes a cluster of small rooms and a series of larger open areas, separated by corridors and partial walls with door openings. This heterogeneous topology creates a mix of LOS and NLOS conditions between anchor pairs, with wall intersection counts ranging from zero (clear LOS) to 3 or more (multiwall NLOS), challenging both autocalibration algorithms. This floor plan is representative of a medium-sized commercial office building, selected to provide a realistic range of LOS and NLOS propagation conditions without resorting to adversarial geometry. A representative deployment within this environment is illustrated in Fig. 2.

2) UWB Radio and Ranging Parameters: The UWB link budget follows (2) with the parameters listed in Table II. The log-distance path loss model in (2) is parameterized using the wall penetration losses listed in Table II.

Range measurements are generated according to (3): each connected link receives a measured range equal to the true Euclidean distance plus a positive NLOS bias (proportional to the number of wall crossings) and an additive zero-mean Gaussian noise term whose standard deviation increases by σ_{wall} per wall crossing beyond LOS. A link is included in the measurement graph if and only if the received power computed by (2) exceeds

TABLE II
UWB SIMULATION PARAMETERS

Parameter	Symbol	Value
Carrier frequency	f_c	6.5 GHz
Transmit power	P_{tx}	-10 dBm
Path loss at $d_0 = 1$ m	PL_0	43.2 dB
Path loss exponent	n	1.8
Rx sensitivity	P_{sens}	-93 dBm
Internal wall loss	L_{int}	5 dB
External wall loss	L_{ext}	12 dB
LOS range noise std	σ_{LOS}	0.20 m
NLOS range noise std	σ_{NLOS}	0.30 m
Per-wall noise increment	σ_{wall}	0.01 m
NLOS bias (internal wall)	b_{int}	0.01 m
NLOS bias (external wall)	b_{ext}	0.02 m
Known-anchor noise std	σ_{KA}	0.10 m
Height measurement std	σ_Z	0.10 m

P_{sens}; links below threshold are recorded as missing entries (NaN) and are unavailable to all algorithms.

Fig. 2 illustrates a representative anchor deployment with $N = 15$ nodes positioned throughout the building at heights between 0.2 and 2.4 m. Connected links are drawn as colored dotted lines, color-coded by the received power at the receiver. Unconnected links below the sensitivity threshold are shown as black solid lines. It is thereby shown that links blocked by multiple external or internal walls fall below -93 dBm and are absent from the ranging graph.

3) Monte Carlo Framework: For the standalone evaluations in Sections V-B and V-C, three parameter sweeps are defined as detailed at the opening of Section V. In each trial the simulator is reinvoked with fresh random anchor positions and a new random noise realization, but all radio parameters remain fixed at the values in Table II. Identical random seeds are used across both standalone algorithms so that performance differences reflect solely the algorithm rather than the deployment geometry. The resulting measurement matrices ($\widetilde{\mathbf{D}}$, **RSSI**, and $\mathbf{W}$) are passed directly to each algorithm without any preprocessing or outlier removal, reflecting realistic deployment conditions where no a priori knowledge of the channel is available.

The standalone and paired-anchor autocalibration algorithms are evaluated via three independent Monte Carlo sweeps, each comprising 100 trials with identical random seeds across methods, enabling direct numerical comparison. In every trial, N anchors are placed at uniformly random positions (minimum separation $d_{\min} = 3.5$ m, heights $\in [0.2, 2.8]$ m) and n_{known} of them are selected at random as reference nodes; a full UWB measurement set is synthesized by the simulator described in Section III. Performance is reported as the median RMSE and per-axis MAE over unknown anchors. Both the *3-D mode* (heights estimated by IDW of known neighbors) and the *semi-3D mode* (heights known to within $\sigma_Z = 0.10$ m) are evaluated throughout. Three sweeps are defined

1) *Sweep A:* $n_{\text{known}} \in \{3 - 8\}$, $N = 20$, $\sigma_{\text{LOS}} = 0.20$ m.
2) *Sweep B:* $\sigma_{\text{LOS}} \in \{0.05 - 0.50\}$ m, $N = 20$, $n_{\text{known}} = 5$.

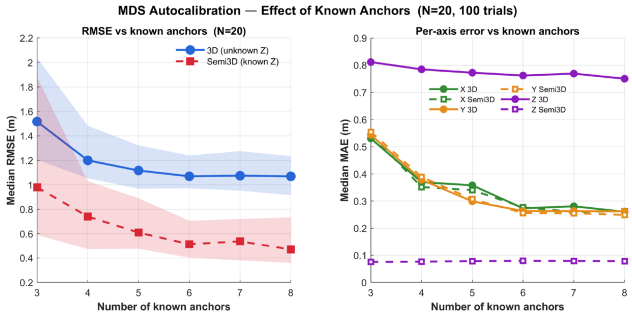


Fig. 3. MDS: RMSE (left) and per-axis MAE (right) versus n_{known}, $N = 20$ and $\sigma_{\text{LOS}} = 0.20$ m. Shaded bands: IQR over 100 trials.

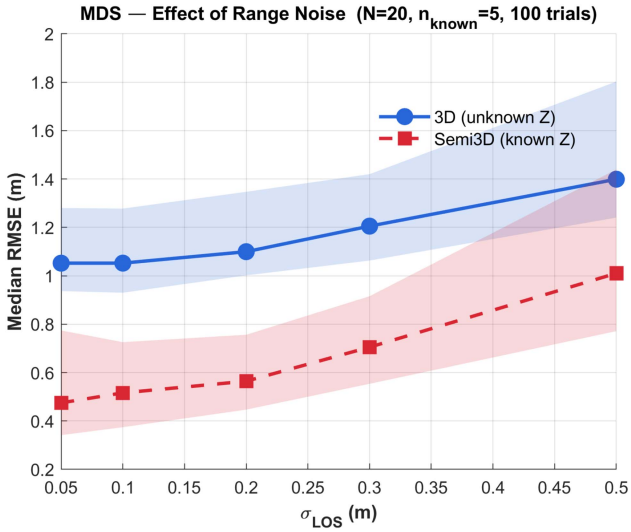


Fig. 4. MDS: RMSE versus σ_{LOS}, $N = 20$ and $n_{\text{known}} = 5$. Shaded bands: IQR over 100 trials.

3) *Sweep C:* $N \in \{8 - 20\}$, $n_{\text{known}} = 4$, $\sigma_{\text{LOS}} = 0.20$ m.

B. MDS Autocalibration

1) Effect of Known-Anchor Count (Sweep A): Fig. 3 shows RMSE and per-axis MAE. In 3-D mode, RMSE falls from 1.52 m at $n_{\text{known}} = 3$ to 1.07 m at $n_{\text{known}} = 6$ (30%), then plateaus—additional reference anchors provide negligible benefit once the Procrustes alignment is sufficiently constrained. Semi-3D mode improves more substantially, from 0.98 to 0.47 m (52%) and continues declining through $n_{\text{known}} = 8$. The persistent gap between modes is explained by the per-axis breakdown: Z MAE in 3-D mode is constant at 0.75–0.81 m regardless of n_{known}, confirming that IDW height estimation is the primary accuracy bottleneck. Semi-3D Z MAE is ≈ 0.08 m (the measurement noise floor σ_Z). Horizontal errors (X and Y) decrease monotonically in both modes. A practical operating point is $n_{\text{known}} = 5$: it captures most of the attainable accuracy (1.12 m/0.61 m in 3-D/semi-3D) with diminishing returns beyond it.

2) Sensitivity to Range Noise (Sweep B): Both modes exhibit monotonically increasing RMSE with noise (see Fig. 4), but at different rates. The 3-D mode rises from 1.05 to 1.40 m (+33%): a modest increase because the height estimation floor (≈ 0.75 m Z MAE) already dominates and masks the effect of

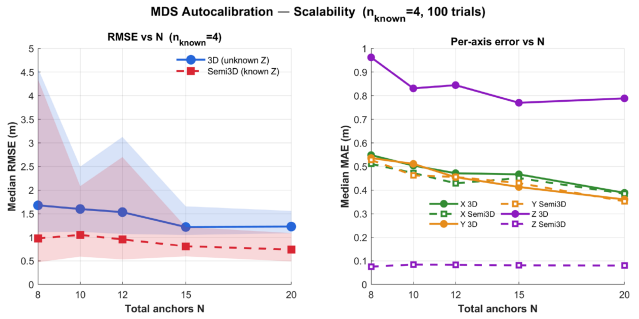


Fig. 5. MDS: RMSE (left, IQR bands) and per-axis MAE (right) versus N, $n_{\text{known}} = 4$ and $\sigma_{\text{LOS}} = 0.20$ m.

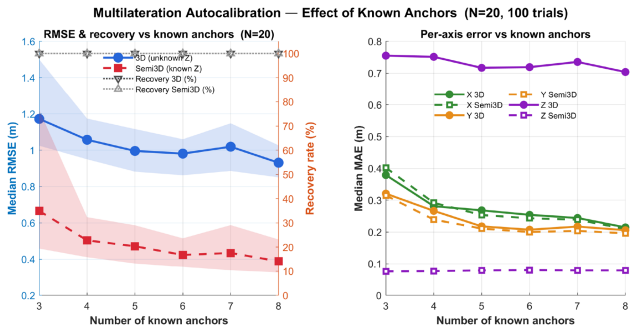


Fig. 6. Multilateration: RMSE and recovery (left, dual y-axis) and per-axis MAE (right) versus n_{known}, $N = 20$ and $\sigma_{\text{LOS}} = 0.20$ m.

ranging noise. Semi-3D starts considerably lower (0.48 m) but climbs more steeply to 1.01 m (+113%), as it is more directly exposed to horizontal ranging errors once height uncertainty is removed. Interquartile range (IQR) bands widen substantially at $\sigma_{\text{LOS}} \geq 0.30$ m for Semi-3D, while 3-D maintains a relatively stable spread.

3) Scalability (Sweep C): Counterintuitively, RMSE *improves* as N grows (see Fig. 5). The 3-D RMSE decreases from 1.68 m at $N = 8$ to 1.23 m at $N = 20$ and semi-3D from 0.98 to 0.74 m. Larger networks produce denser measurement graphs, reducing Floyd–Warshall path inflation globally. However, variance is strongly N-dependent: IQR half-widths of 1.73 m (3-D) and 1.94 m (semi-3D) at $N = 8$ collapse to 0.24 m at $N = 20$, indicating that MDS is fragile under poorly conditioned small graphs. The anomalous peak at $N = 12$ (IQR ± 1.03 m) reflects ill-conditioned topologies where Floyd–Warshall (FW) gap-fill introduces significant distortion. Networks of $N \leq 12$ should be treated with caution.

C. Multilateration Autocalibration

1) Effect of Known-Anchor Count (Sweep A): The cascade achieves 100% recovery in every trial across all n_{known} values: the dense connectivity of a 20-anchor network always provides each anchor with $\geq M_{\min} = 3$ reachable references. RMSE decreases from 1.17 to 0.93 m in 3-D and from 0.67 to 0.39 m in semi-3D, as shown in Fig. 6. Multilateration outperforms MDS at every operating point—at $n_{\text{known}} = 3$ by 23% in 3-D and 32% in semi-3D—because it uses *measured* distances rather than multihop Floyd–Warshall paths. The per-axis pattern is

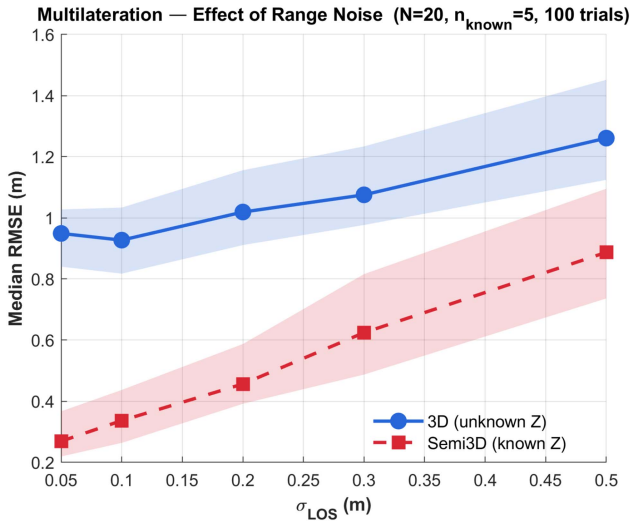


Fig. 7. Multilateration: RMSE versus σ_{LOS}, $N = 20$ and $n_{\text{known}} = 5$. Shaded bands: IQR over 100 trials.

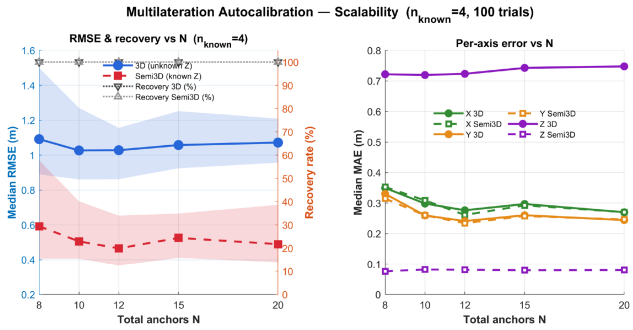


Fig. 8. Multilateration: RMSE and recovery (left, dual y-axis) and per-axis MAE (right) versus N, $n_{\text{known}} = 4$ and $\sigma_{\text{LOS}} = 0.20$ m.

qualitatively identical to MDS: Z MAE constant at 0.70–0.76 m in 3-D (IDW floor), ≈ 0.08 m in semi-3D; horizontal errors decrease monotonically with n_{known}.

2) Sensitivity to Range Noise (Sweep B): Recovery remains 100% at all noise levels. The 3-D RMSE rises from 0.95 to 1.26 m and semi-3D from 0.27 to 0.89 m, as shown in Fig. 7. At $\sigma_{\text{LOS}} = 0.05$ m, semi-3D multilateration (0.27 m) is 43% better than semi-3D MDS (0.475 m), reflecting the complete elimination of FW distortion. This advantage narrows to 12% at $\sigma_{\text{LOS}} = 0.50$ m as both methods become noise-dominated. The 3-D mode's flat response is again explained by the IDW height floor masking horizontal ranging noise.

3) Scalability (Sweep C): Unlike MDS, multilateration RMSE is nearly flat with N: 3D varies between 0.93 and 1.09 m with no consistent trend; semi-3D improves modestly from 0.59 to 0.49 m (see Fig. 8). This is a structural property: each anchor is estimated locally, so additional network anchors do not improve the local geometric sub-problem.

D. Paired-Anchor Autocalibration Evaluation

The paired-anchor autocalibration algorithms described in Section IV-C are evaluated via the same three Monte Carlo sweeps used for the standalone methods, over 100 independent

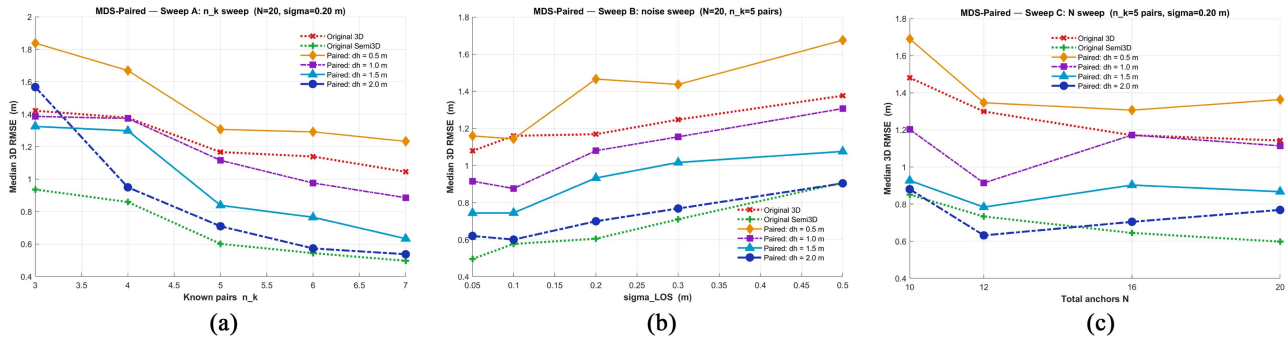


Fig. 9. MDS-Paired autocalibration: Median 3-D RMSE for $\Delta h \in \{0.5, 1.0, 1.5, 2.0\}$ m (colored lines) compared against the standalone MDS 3-D baseline (red dotted) and semi-3D upper bound (green dotted), over 100 Monte Carlo trials. (a) Sweep A: n_k sweep ($N = 20$ and $\sigma_{\text{LOS}} = 0.20$ m). (b) Sweep B: noise sweep ($N = 20$ and $n_k = 5$ pairs). (c) Sweep C: N sweep ($n_k = 5$ pairs and $\sigma_{\text{LOS}} = 0.20$ m).

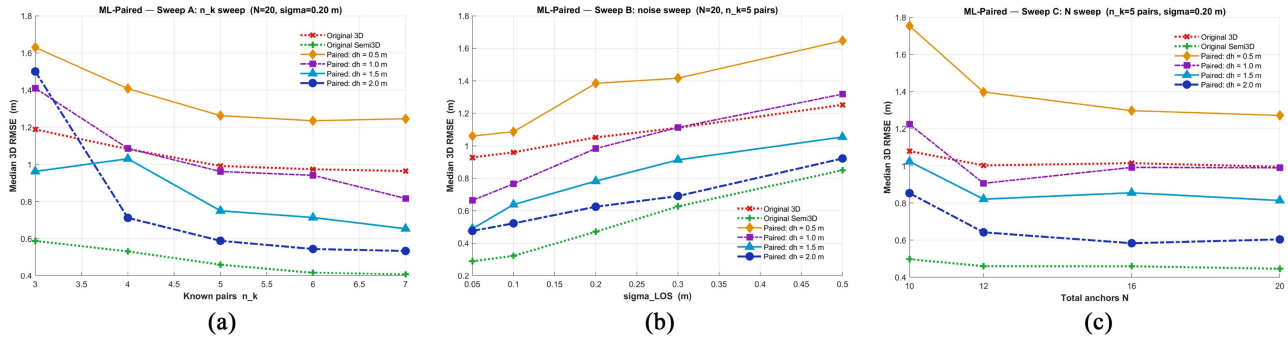


Fig. 10. ML-Paired autocalibration: Median 3-D RMSE for $\Delta h \in \{0.5, 1.0, 1.5, 2.0\}$ m (colored lines) compared against the standalone ML 3-D baseline (red dotted) and semi-3D upper bound (green dotted), over 100 Monte Carlo trials. (a) *Sweep A:* n_k sweep ($N = 20$ and $\sigma_{\text{LOS}} = 0.20$ m). (b) *Sweep B:* noise sweep ($N = 20$ and $n_k = 5$ pairs). (c) *Sweep C:* N sweep ($n_k = 5$ pairs and $\sigma_{\text{LOS}} = 0.20$ m).

trials. In each trial, N anchors are placed in $M = N/2$ vertically separated pairs at randomly chosen XY positions. Then, n_k pairs are selected as calibrated references and a full UWB measurement set is generated by the physics-based simulator. Four vertical separations $\Delta h \in \{0.5, 1.0, 1.5, 2.0\}$ m are evaluated in parallel for each trial configuration. Performance is reported as the median 3-D RMSE over the unknown pairs. The standalone 3-D and semi-3D results from Sections V-B and V-C are overlaid as reference baselines.

1) Effect of Known-Pair Count (Sweep A): Sweep A fixes $N = 20$ and $\sigma_{\text{LOS}} = 0.20$ m while varying $n_k \in \{3, 4, 5, 6, 7\}$. Results for MDS-Paired and Multilateration (ML)-Paired are shown in Figs. 9(a) and 10(a) respectively.

MDS-paired: The vertical separation Δh is the dominant performance parameter. At $\Delta h = 0.5$ m the paired approach is uniformly worse than the standalone 3-D baseline across all values of n_k, with RMSE ranging from 1.84 m at $n_k = 3$ to 1.23 m at $n_k = 7$. This confirms the theoretical prediction of (30): with small Δh the differential estimator amplifies ranging noise and the Z improvement is insufficient to offset the loss of XY reference diversity due to the halved number of anchor positions. Increasing the separation to $\Delta h = 1.0$ m produces a crossover at $n_k = 5$, above which the paired algorithm outperforms the standalone 3-D mode (1.12 versus 1.17 m at $n_k = 5$, improving to 0.89 versus 1.05 m at $n_k = 7$). At $\Delta h = 1.5$ m, the crossover moves to $n_k = 4$ and the gap widens substantially at higher known-pair counts, reaching 0.63 m at $n_k = 7$—a 40% reduction over the 3-D baseline. The largest separation $\Delta h = 2.0$ m is more variable at low n_k (1.57 m at $n_k = 3$ versus 1.42 m for Orig-3D), but converges to the lowest RMSE of any configuration at $n_k \geq 5$, achieving 0.54 m at $n_k = 7$—within 8% of the semi-3D reference (0.50 m) and 48% below the standalone 3-D result.

ML-paired: The ML-paired algorithm follows the same qualitative trends with Δh playing the same dominant role. The standalone ML-3D baseline is lower than its MDS counterpart throughout (1.19–0.96 m versus 1.42–1.05 m), reflecting the cascade's superior XY accuracy. At $\Delta h = 2.0$ m and small n_k the paired approach is again unstable (1.50 m at $n_k = 3$ versus 1.19 m Orig-ML-3D), indicating that the cascade requires at least four to five reference pairs to initialize reliably. From $n_k = 4$ onward, $\Delta h = 1.5$ and 2.0 m both beat the standalone baseline, with $\Delta h = 2.0$ m achieving 0.53 m at $n_k = 7$—45% below the 0.96 m baseline and approaching the semi-3D floor of 0.41 m.

2) Noise Robustness (Sweep B): Sweep B fixes $N = 20$ and $n_k = 5$ while varying $\sigma_{\text{LOS}} \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$ m. Results are shown in Figs. 9(b) and 10(b).

MDS-paired: The $\Delta h = 2.0$ m curve is the most noise-robust configuration, ranging from 0.63 m at $\sigma = 0.05$ to 0.91 m at $\sigma = 0.50$ m—a monotonically increasing but consistently low-error response that remains well below the standalone 3-D baseline (1.09–1.38 m over the same range). The relative improvement over the standalone 3-D mode is 34%–42% and grows modestly with noise level, indicating that the height-decoupling property

of the differential estimator (27) is effective and increasingly advantageous as ranging noise increases. At $\Delta h = 0.5$ m the paired algorithm is again worse than standalone 3D.

ML-paired: A very similar pattern emerges for ML-paired. At $\Delta h = 2.0$ m, the RMSE spans 0.48–0.93 m compared to 0.93–1.25 m for the standalone ML-3D baseline, an improvement of 26%–48%. The $\Delta h = 1.5$ m curve achieves RMSE below 1.10 m at all noise levels, consistently outperforming the standalone 3-D mode. The $\Delta h = 0.5$ m curve reaches 1.39 m at $\sigma_{\text{LOS}} = 0.20$ m and rises to 1.65 m at $\sigma_{\text{LOS}} = 0.50$ m, confirming that the small-separation regime is detrimental at all noise levels.

3) Network Size Sensitivity (Sweep C): Sweep C fixes $n_k = \min(5, M-1)$ and $\sigma_{\text{LOS}} = 0.20$ m while varying $N \in \{10, 12, 16, 20\}$. Results are shown in Figs. 9(c) and 10(c).

MDS-paired: At all network sizes, $\Delta h = 2.0$ m yields the lowest RMSE. The largest relative gain occurs at $N = 12$, where $\Delta h = 2.0$ m achieves 0.63 m against the standalone 3-D result of 1.30 m—a 51% reduction. The trend is nonmonotonic at larger N: RMSE increases slightly from $N = 12$ to $N = 16$ before stabilizing at $N = 20$, partly because n_k saturates at 5 while the total number of unknown pairs grows, diluting the relative reference density. The paired method nonetheless closes the gap between standalone 3-D and semi-3D substantially for all $N \geq 12$.

ML-paired: The ML-Paired results mirror the MDS-Paired trends. At $N = 12$, $\Delta h = 2.0$ m achieves 0.64 versus 1.00 m for standalone ML-3D (36% improvement). At $N = 20$, the gap is 0.60 versus 1.00 m (40% improvement). Unlike MDS-Paired, the ML-paired RMSE at $\Delta h = 2.0$ m decreases monotonically with N, reflecting the cascade's natural benefit from a denser connectivity graph as more anchors are added. Both methods with $\Delta h = 2.0$ m remain below 0.65 m for $N \geq 12$, substantially outperforming their respective 3-D baselines.

4) NLOS Severity Sensitivity (Sweep D): In practice, UWB anchor-to-anchor ranges are affected not only by zero-mean Gaussian noise but also by a systematic positive bias caused by multipath propagation and NLOS wall crossings. The parameter b_{int}, which appears in the physics-based channel model described in Section III [(4) and (5)], controls the per-internal-wall range bias added to each obstructed link; the per-external-wall counterpart is set to $b_{\text{ext}} = 2\, b_{\text{int}}$ throughout, consistent with the higher attenuation of external masonry walls. Because b_{int} is the single parameter that governs the severity of systematic NLOS distortion in the simulated building, sweeping it provides a direct measure of how the paired-anchor framework degrades as NLOS conditions worsen.

The closed-form cross-pair estimators in (4) and (5) exploit the differential geometry between the low and high anchors of each pair. When both cross-links between two pairs traverse the same obstructing walls, the NLOS bias terms cancel in the differencing operation, leaving the estimator unaffected. Residual sensitivity arises only when the two cross-links experience *asymmetric* wall blockage, i.e., when only one of them is NLOS-affected, so that the cancelation is incomplete. Sweep D is designed to make this mechanism visible experimentally. The sweep fixes $N = 20$ anchors, $n_k = 5$ known pairs, and $\Delta h = 2.0$ m, and varies b_{int}

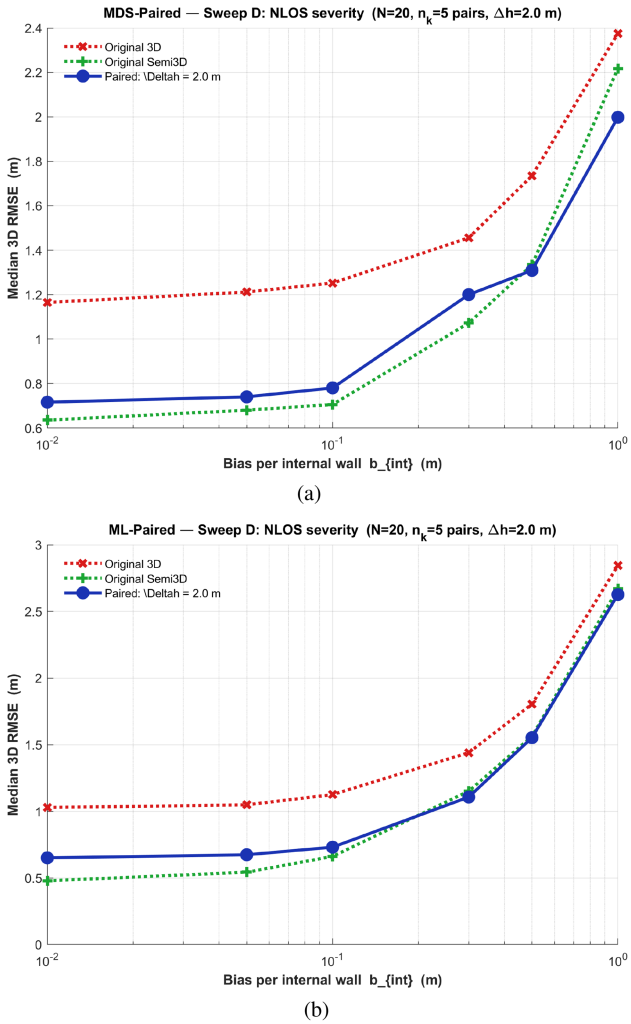


Fig. 11. 3-D RMSE versus per-internal-wall NLOS bias b_{int} for all Δh configurations. Top: MDS-paired. Bottom: ML-paired. Fixed parameters: $N = 20$, $n_k = 5$, and $\sigma_{\text{LOS}} = 0.20$ m. (a) MDS-paired. (b) ML-paired.

over $\{0.01, 0.05, 0.10, 0.30, 0.50, 1.00\}$ m. The LOS range noise is held at $\sigma_{\text{LOS}} = 0.20$ m. The lower end of the sweep ($b_{\text{int}} \leq 0.10$ m) corresponds to light NLOS typical of modern IEEE 802.15.4z UWB transceivers in furnished office environments; the upper end ($b_{\text{int}} = 1.00$ m) represents heavily obstructed NLOS well beyond what compliant devices are expected to encounter.

Results are presented in Fig. 11 for the MDS-Paired and ML-Paired variants. At practical bias levels ($b_{\text{int}} \leq 0.10$ m), both paired variants retain their full accuracy advantage over the original 3-D baseline, achieving a reduction of approximately 35%–40% in median 3-D RMSE. This confirms that the differential estimator successfully suppresses the dominant common-mode bias component under realistic NLOS conditions. As bias grows, the advantage narrows due to increasing asymmetric NLOS, with the three methods converging at extreme values ($b_{\text{int}} \geq 0.50$ m); nonetheless, even at $b_{\text{int}} = 1.00$ m the MDS-Paired variant (2.00 m RMSE) remains below both semi-3D (2.21 m) and original 3-D (2.37 m). These results confirm that the paired-anchor architecture provides meaningful

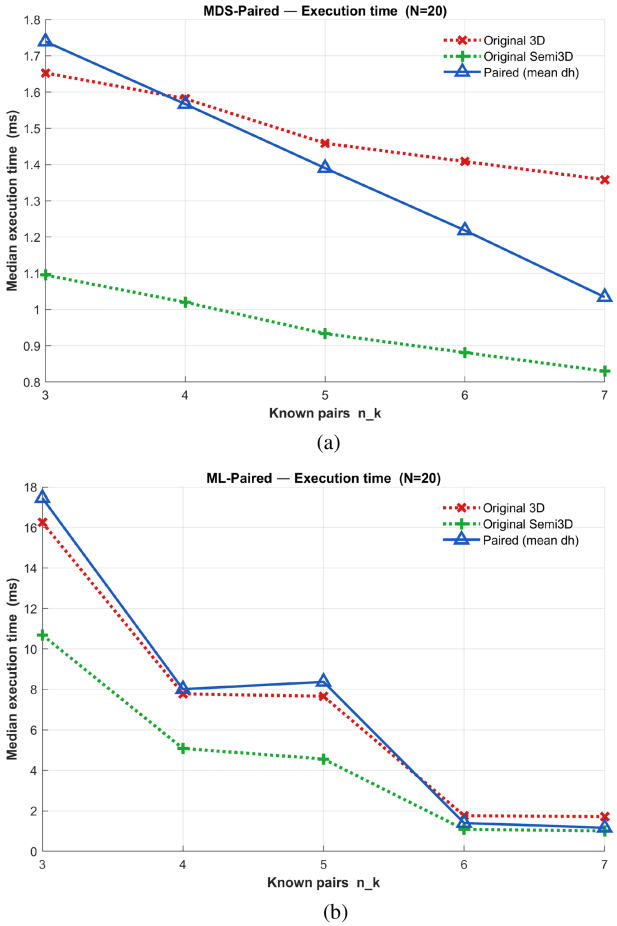


Fig. 12. Median per-trial execution time versus known-pair count n_k. The paired curve is the mean over $\Delta h \in \{0.5,\ 1.0,\ 1.5,\ 2.0\}$ m. (a) MDS-paired. (b) ML-paired.

NLOS resilience across the full range of bias levels likely to be encountered in practice, without requiring explicit NLOS identification or mitigation.

5) Computational Overhead: Fig. 12 compares the median per-trial execution times of the standalone 3-D, standalone semi-3D and paired algorithms for both modes. The paired curve reports the mean over all four Δh values, as execution time is insensitive to Δh.

For MDS-Paired [see Fig. 12(a)], execution times range from 1.04 to 1.75 ms and closely track the standalone 3-D curve, confirming that the postprocessing step (CROSSPAIRDIST, pair averaging, trimmed-median Z) contributes negligible overhead relative to the MDS backbone. The slight decrease with n_k reflects the reduced number of unknown anchors that the pivot-LS refinement must process. For ML-paired [see Fig. 12(b)], execution times are substantially higher at $n_k = 3$ (≈16–18 ms) due to cascade instability under sparse references, but converge to 1–2 ms at $n_k \geq 6$ — on a par with the standalone ML-3D. The steep decline for all three ML curves reflects the cascade's ability to resolve unknown anchors in fewer iterations when more references are available. In both cases the paired approach introduces no material computational penalty over the corresponding standalone algorithm.

TABLE III
3-D RMSE (M) AT $N = 20$, $n_k = 5$, AND $\sigma_{\text{LOS}} = 0.20$ M; LOS CONDITIONS ($b_{\text{INT}} = 0$)

Method	Orig. 3-D	Orig. semi-3D	$\Delta h = 1.5$ m	$\Delta h = 2.0$ m
MDS-Paired	1.167	0.602	0.839	0.710
ML-Paired	0.991	0.460	0.750	0.588

NLOS sensitivity is characterized separately in sweep D (see Fig. 11).

6) Summary: Table III summarizes the best paired configurations at the common operating point $N = 20$, $n_k = 5$, and $\sigma_{\text{LOS}} = 0.20$ m alongside the corresponding standalone and semi-3D baselines.

The paired approach with $\Delta h = 2.0$ m achieves the best results for both backbone algorithms, reducing the 3-D RMSE by 39% (MDS) and 41% (ML) relative to the standalone 3-D baseline at the $n_k = 5$ operating point and by approximately 49% (MDS) and 45% (ML) at $n_k = 7$. The improvement increases monotonically with Δh, confirming that the noise variance of the differential height estimator (30)—which decreases as $1/\Delta h^2$—is the principal driver of performance. The minimum effective separation is approximately $\Delta h = 1.0$ m for MDS-paired and $\Delta h = 1.5$ m for ML-paired with $n_k \geq 5$ pairs. With a practical ceiling height of 3.0 m and a low-anchor height range of $[0.2,\ 1.0]$ m, a separation of $\Delta h = 2.0$ m is readily achievable and is recommended as the deployment target. Sweep D further demonstrates that the paired-anchor framework is resilient to realistic NLOS conditions: at practical per-internal-wall bias levels ($b_{\text{int}} \leq 0.10$ m), the 35% –40% RMSE advantage over the standalone 3-D baseline is fully preserved, owing to the common-mode bias cancelation inherent in the cross-pair differential estimators (27). Even under extreme multipath distortion ($b_{\text{int}} = 1.00$ m), MDS-Paired achieves 2.00 m RMSE, remaining below both the semi-3D (2.21 m) and standalone 3D (2.37 m) baselines, without requiring explicit NLOS identification or mitigation.

7) Single-Trial 3-D Visualization: To complement the statistical Monte Carlo results, Fig. 13 presents a single representative trial of the paired-anchor deployment for both MDS-paired (left) and ML-paired (right). The trial uses $N = 40$ individual anchors ($M = 20$ pairs) with $\Delta h = 1.5$ m, $\sigma_{\text{KA}} = 0.10$ m, $\sigma_{\text{LOS}} = 0.20$ m, $\sigma_{\text{NLOS}} = 0.30$ m, and $n_k = 5$ randomly selected known pairs (green diamonds). Unknown pairs are shown as black circles at their true low-anchor positions, with the vertical bar to the corresponding high anchor rendered in grey. Error vectors (dashed grey lines) connect each true position to its estimate and each label reports the pair index and displacement in meters.

Both algorithms recover the spatial layout of the network with good visual fidelity. The majority of estimates cluster tightly around their true positions, with error vectors short enough that estimates and true positions often overlap, confirming the sub-meter accuracy reported in Sections V-D. ML-Paired achieves a lower per-trial RMSE (0.588 versus 0.650 m), consistent with the statistical advantage of the cascade's greedy nearest neighbor resolution over the global MDS embedding for this network size and noise level.

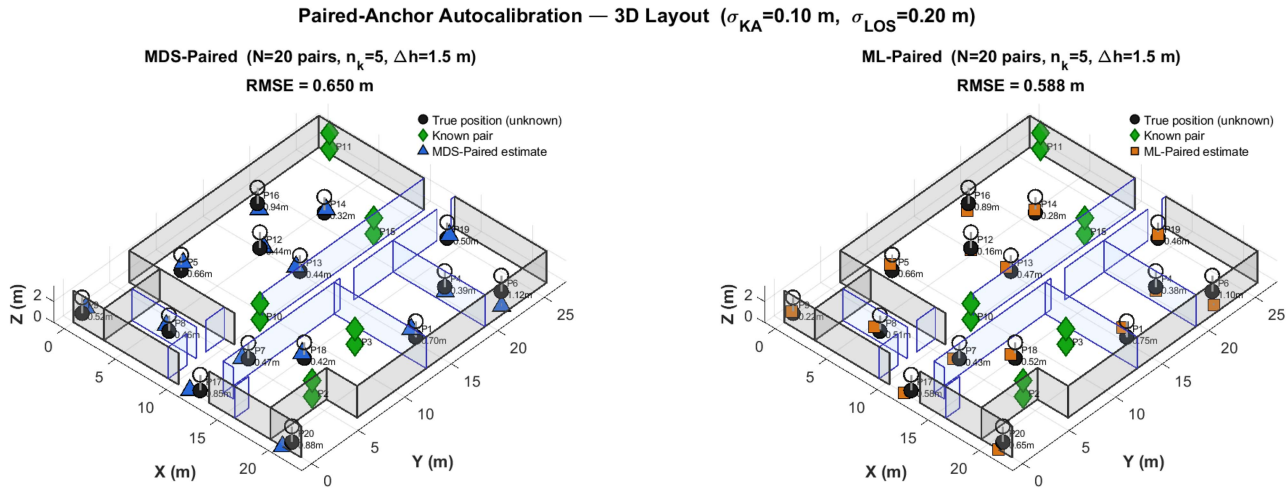


Fig. 13. Single-trial 3-D visualization of paired-anchor autocalibration ($N = 40$ anchors, $M = 20$ pairs, $n_k = 5$, $\Delta h = 1.5$ m, $\sigma_{\text{KA}} = 0.10$ m, and $\sigma_{\text{LOS}} = 0.20$ m). Left: MDS-Paired (blue triangles), RMSE = 0.650 m. Right: ML-Paired (orange squares) and RMSE = 0.588 m. Green diamonds: known pairs. Black circles: true unknown positions. Dashed lines: error vectors from true to estimated position.

E. Discussion

1) Summary of Key Findings: This work has developed and evaluated two lightweight anchor autocalibration algorithms—MDS-based and multilateration-based—in both full 3-D and semi-3D operating modes and has proposed a structured paired-anchor extension that substantially reduces height estimation error through a closed-form differential formula. The results across four parameter sweeps yield several clear conclusions.

First, *height estimation is the dominant accuracy bottleneck in 3-D mode.* In every configuration evaluated, the Z MAE remains between 0.70 and 0.82 m regardless of the number of known anchors, ranging noise level, or network size. The horizontal components (X and Y) respond readily to more known anchors and better connectivity, but the 3-D RMSE is dominated throughout by the vertical term.

Second, *multilateration consistently outperforms MDS in XY accuracy*. At $n_{\text{known}} = 5$, $N = 20$, and $\sigma_{\text{LOS}} = 0.20$ m, multilateration achieves 0.99 m 3-D RMSE compared to 1.17 m for MDS—a 15% advantage that persists across all sweep configurations. This arises because multilateration uses direct measured distances whereas MDS relies on Floyd–Warshall multihop paths whose accumulated overestimation, despite shrinkage correction, introduces global distortion. However, MDS has a structural robustness advantage: it provides a simultaneous global embedding that is less sensitive to cascade ordering effects and it is substantially more stable at small N where multilateration recovery rates degrade.

Third, *the paired-anchor deployment effectively decouples height estimation from ranging noise.* At $\Delta h = 2.0$ m and $n_k = 5$ known pairs, the paired approach reduces 3-D RMSE by 39% (MDS) and 41% (multilateration) relative to the standalone 3-D baselines. At $n_k = 7$, the reduction reaches approximately 49% (MDS) and 45% (multilateration), with the best results approaching within 8%–10% of the semi-3D ceiling. The $1/\Delta h^2$ dependence of the differential estimator variance (30) explains why larger vertical separations are so effective: doubling Δh from 1.0 to 2.0 m reduces the height estimation variance by a factor of four, yielding improvements that outweigh the XY reference density reduction caused by halving the number of distinct anchor positions.

Fourth, *the paired-anchor differential estimator provides inherent resilience to realistic NLOS conditions.* Sweep D varies the per-internal-wall ranging bias b_{int} over $[0.01, 1.00]$ m and demonstrates that at bias levels typical of IEEE 802.15.4z UWB transceivers in furnished office environments ($b_{\text{int}} \leq 0.10$ m), the 35%–40% RMSE advantage of the paired approach over the standalone 3-D baseline is fully preserved. This robustness arises naturally from the common-mode cancelation property of the cross-pair differencing in (27): when both cross-links between two pairs traverse the same obstructing walls, the NLOS bias terms cancel exactly, leaving the estimator unaffected. Residual sensitivity is limited to cases of asymmetric wall blockage and, even at the extreme value $b_{\text{int}} = 1.00$ m, the MDS-paired variant (2.00 m RMSE) remains below both the semi-3D (2.21 m) and standalone 3-D (2.37 m) baselines without requiring any explicit NLOS identification or mitigation.

2) Comparison With the Literature: Placing these results in the context of closely related published work requires careful accounting for differences in dimensionality, environment and measurement methodology.

a) Comparison with Herbruggen et al. [6]: The most directly comparable multihop UWB self-calibration system is that of Herbruggen et al. [6], who reported an MAE of 21.6 cm in a real-world industrial environment (25×11 m^2, 18 nodes, metallic NLOS obstacles) using a three-stage algorithm combining linear least squares (LLS) initialization, L-BFGS-B global optimization and an optional Kalman-filter refinement with a mobile node. This figure is substantially lower than the XY errors reported in the present work. However, several fundamental differences temper direct comparison. The most consequential is *dimensionality:* Herbruggen et al. [6] solved a 2-D problem, with all anchors mounted at a fixed height of 1.60 m on tripods.

In the present work, anchor heights span $[0.2, 2.8]$ m and must be estimated jointly with horizontal coordinates, creating an intrinsically harder problem. A second consideration is *the optional mobile node in* [6, Stage 3], which collects additional measurements from a handheld tag walked through the deployment area. Without this stage, the algorithm in [6] achieves 28.2 cm MAE (Stage 2 only), which is a fairer comparator to the present work's anchor-only approach. The proposed paired algorithm, which adds no mobile node and no additional sensing, achieves 2-D XY RMSE values below 0.50 m at $n_k \geq 5$ in the horizontal plane, suggesting comparable XY accuracy relative to [6, Stage 2] when evaluated on a 2-D basis—although a definitive quantitative comparison would require evaluation on real hardware in a shared environment. A further structural distinction is *computational complexity*. [6, Stage 2] uses L-BFGS-B global optimization, an iterative nonlinear solver whose convergence depends on the cost function landscape and can require many gradient evaluations in large networks. The proposed MDS and multilateration algorithms are noniterative in their core stages: MDS costs $O(N^3)$ for a single eigendecomposition, and multilateration costs $O(M_{\max}^2\, N)$ for N pivot-LS solves. Both are fully deterministic given the measurement inputs, with no optimization hyperparameters to tune.

b) Comparison with Ridolfi et al. [5]: Ridolfi et al. [5] reported the best case accuracy of 2.4 cm average error (95th percentile: 5.7 cm) for UWB anchor self-calibration in realistic NLOS conditions, achieved through a combination of iterative gradient descent, machine learning classification of channel impulse response (CIR) signals for NLOS detection and adaptive physical layer (PHY) configuration that selects per-link physical settings to maximize ranging quality. This accuracy level is well beyond what the present work achieves. However, the comparison is not straightforward. The accuracy of [5] is conditioned on access to the full CIR signal from each UWB link and on the availability of a convolutional neural network trained on measurements from the specific environment and hardware platform. The present work intentionally targets *infrastructure-only autocalibration with no learning components:* the algorithms operate on scalar ranging measurements only, require no CIR data and carry no trained models that would need to be retrained when the hardware platform or environment changes. This makes the proposed approach significantly more portable and easier to deploy in new environments without a prior data collection phase. The 2.4 cm accuracy of [5] should therefore be understood as an upper bound on what is achievable when full CIR processing and adaptive PHY are available, rather than as a baseline for lightweight anchor-only autocalibration. A second consideration is again *dimensionality:* the evaluation in [5] is 2-D, with anchors in a fixed-height deployment, whereas the present work targets 3-D environments with variable anchor heights. Furthermore, the environment in [5] contains up to eight anchors in a relatively small volume, whereas the present work targets networks of 15–20 anchors in a 22.3 m $\times$ 26.3 m $\times$ 3.0 m building with multiroom NLOS structure.

c) Comparison with Schwarzbach and Jung [7]: The probabilistic grid-based approach of Schwarzbach and Jung [7] is the most recent comparable work, operating in 2-D on a fully meshed 12-node network in a 44 $\times$ 30 m^2 industrial hall. Under LOS initialization, their probabilistic grid-based positioning (PGP) achieves median errors of 0.34–0.68 m, comparable to the horizontal RMSE of the present work's algorithms in their semi-3D modes (0.39–0.61 m). The PGP method's key advantage over deterministic approaches is its robustness under NLOS-contaminated initialization, where it maintains 0.62–0.99 m median accuracy while closed-form baselines collapse to 1.48–2.43 m. The present work does not evaluate NLOS-contaminated reference anchor initialization—the known anchor positions are assumed to be available with bounded Gaussian uncertainty $\sigma_{\mathrm{KA}} = 0.10$ m—so this robustness dimension cannot be directly compared. However, the NLOS robustness of PGP comes at a computational cost of $\approx 2.3 \times 10^7$ likelihood evaluations per epoch $[O(\tau^2\, M^2\, N^2)]$, whereas the present algorithms complete in 1–2 ms per trial on a standard workstation, a difference of several orders of magnitude that becomes significant for large-scale or battery-constrained deployments.

3) Limitations and Scope: The present work is best understood as a *proof-of-principle study* for the paired-anchor autocalibration concept: its primary contribution is to demonstrate, through systematic Monte Carlo simulation, that a structured vertical deployment combined with a closed-form differential height estimator can substantially close the gap between lightweight 3-D autocalibration and the semi-3D performance ceiling. Several limitations of this simulation-based validation must be acknowledged before the approach can be recommended for deployment.

The most significant limitation is the *simulation-only evaluation*. No real UWB hardware was used to validate the algorithms. The physics-based simulator captures the dominant propagation effects—Gaussian ranging noise, NLOS wall-crossing bias, and the building's multiroom geometry—but real UWB transceivers exhibit additional systematic errors, including antenna directional gain patterns, clock drift, hardware-specific ranging biases that vary with received power, and intermittent packet loss due to fading, which are not modeled. It is therefore expected that absolute RMSE values observed on real hardware will be higher than those reported here. Experimental validation on real hardware in a realistic indoor environment is the most pressing next step, and one that would also enable a direct, fair comparison with the experimentally obtained results of [6].

A second limitation is the *single building geometry used throughout*. All Monte Carlo evaluations use the same 22.3 m $\times$ 26.3 m multiroom floor plan. While anchor positions and known-anchor subsets are randomized across trials, the wall layout and NLOS pattern are fixed. Buildings with different aspect ratios, more irregular room layouts, or higher wall densities may produce different connectivity graphs and different asymmetric-NLOS incidence rates for the cross-pair links, affecting the robustness results reported in Sweep D. Evaluation across a broader range of building geometries remains an open direction.

For the paired-anchor approach specifically, the practical requirement of installing *two* anchors at each XY position doubles the hardware count relative to a conventional single-layer deployment, which may constrain applicability where cost or physical mounting is limited. In addition, the benefit of large Δh

requires sufficient ceiling height: in low-ceiling environments (below 2.5 m) the achievable separation is correspondingly limited, reducing the height-estimation gain. The minimum effective separations identified in the simulation ($\Delta h \approx 1.0$ m for MDS-paired and $\Delta h \approx 1.5$ m for ML-paired with $n_k \geq 5$) should be treated as deployment guidelines to be reverified on real hardware rather than as universal thresholds.

Notwithstanding these limitations, the simulation results provide a clear and consistent demonstration of the principle: pairing anchors vertically and exploiting the differential geometry of the cross-pair measurements is an effective strategy for decoupling height estimation from horizontal geometry error in 3-D UWB autocalibration.

VI. Conclusion

A lightweight, purely anchor-based 3-D autocalibration framework for sparse multihop UWB networks has been presented, with the primary aim of establishing the proof of principle for a structured paired-anchor deployment as a practical path to closing the accuracy gap between 3-D and semi-3D autocalibration without requiring additional infrastructure or learning components. Two complementary algorithms were developed and evaluated: an MDS-based approach that recovers the global anchor geometry via Floyd–Warshall gap-fill, Procrustes alignment and iterative pivot-LS refinement; and a multilateration-based greedy cascade that propagates the known-anchor set outward through the network. Both were evaluated in full 3-D and semi-3D modes across four Monte Carlo parameter sweeps, achieving 3-D RMSE of 1.17 and 0.99 m, respectively, at the reference configuration ($N = 20, n_{\text{known}} = 5$, and $\sigma_{\text{LOS}} = 0.20$ m). Height estimation was identified as the dominant accuracy bottleneck in 3-D mode, with the IDW-based Z estimator contributing a near-constant ≈ 0.75 m MAE irrespective of network parameters.

To address this bottleneck, a structured *paired-anchor* deployment was introduced in which two anchors share each XY position with a fixed vertical separation Δh. Cross-pair ranging yields a closed-form height estimator that exactly cancels the horizontal interpair distance, making Z-estimation independent of horizontal position error. The noise variance of this estimator scales as $1/\Delta h^2$, providing a clear deployment guideline: maximize the vertical separation within the available ceiling height. At $\Delta h = 2.0$ m and $n_k = 5$ known pairs, the paired approach reduces 3-D RMSE by 39% (MDS) and 41% (multilateration) over the standalone 3-D baselines, reaching 0.54 and 0.53 m at $n_k = 7$—within 10% of the semi-3D performance ceiling—with negligible computational overhead. Sweep D further demonstrates that this advantage is maintained under realistic NLOS conditions ($b_{\text{int}} \leq 0.10$ m typical of IEEE 802.15.4z transceivers in furnished environments), owing to the inherent common-mode bias cancelation of the cross-pair differential estimator.

These simulation results validate the paired-anchor concept and provide a quantitative basis for a physical deployment study. Future work will focus on three directions: 1) experimental validation on real UWB hardware in a representative indoor environment, which is essential to quantify the simulation-to-reality gap, account for hardware-specific ranging biases and packet loss, and enable direct comparison with the experimentally validated results of [6]; 2) incorporation of lightweight NLOS detection via residual-based outlier gating to improve robustness in heavily obstructed deployments; and 3) extension of the differential height estimator to anchor-to-tag measurements, providing additional vertical constraints without requiring a dedicated mobile survey node.

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